

Algebraic Topology 3

Labix

July 11, 2026

Abstract

References:

- Notes on Algebraic Topology by Oscar Randal-Williams:
The first chapter gives a complete treatment of the first three sections of these notes, as well as providing the importance of fibrations on the higher homotopy groups. These notes are highly recommended to understanding the first three sections.
- Algebraic Topology by Allen Hatcher:
A more or less complete dictionary on all topics of these notes. However it is prone to the same problem in the sense that Hatcher's book is rather terse and definitions and parts of some theorems are scattered throughout the paragraphs rather than having a complete statement for reference. Nevertheless it is still the standard reference of the notes, albeit organized in a slightly different way.
- A non-visual proof that higher homotopy groups are abelian by Shintaro Fushida-Hardy:
This short piece of article proves that the higher homotopy groups are abelian in a purely algebraic way. Most geometric visualization of such a proof has the same underlying idea as the algebraic method.

Contents

I	Some Point Set Topology	4
1	The Compact Open Topology	5
1.1	The Mapping Space	5
1.2	The Smash Product and the Pointed Mapping Space	5
2	Suspension and Loopspaces	7
2.1	Suspensions and Reduced Suspensions	7
2.2	Loopspaces and Free Loopspaces	8
2.3	The Suspension-Loopspace Adjunction	8
II	The Higher Homotopy Groups and Their Properties	10
3	The Higher Homotopy Groups	11
3.1	The n th Homotopy Groups	11
3.2	Functorial Properties of Homotopy	14
3.3	The Action of the Fundamental Group	16
3.4	Relative Homotopy Groups	16
3.5	Induced Maps of Relative Homotopy Groups	16
3.6	Long Exact Sequence in Homotopy Groups	17
3.7	Homotopical Connectivity	18
4	Whitehead's Theorem and CW Approximations	19
4.1	Whitehead's Theorem	19
4.2	Cellular Approximations	19
4.3	CW Approximations	20
4.4	Eilenberg-MacLane Spaces	21
5	Properties of the Homotopy Groups	22
5.1	The Homotopy Excision Theorem	22
5.2	Freudenthal Suspension Theorem	22
5.3	The Homotopy Groups of a Fiber Bundle	24
5.4	Weak Equivalences and Homological Isomorphisms	24
5.5	The Hurewicz Homomorphism	24
6	H-Spaces	26
6.1	H-Spaces and the Hopf Algebra Structure on Cohomology	26
6.2	Homotopy Associative H-Space and the Pontryagin Product	27
6.3	Group Like H-Spaces	27
6.4	Co-H-Space	28
6.5	Group Structure on the Set of Maps	28
6.6	Group Isomorphism on the Suspension-Loopspace Adjunction	29
7	The Free Monoid of a Space	31
7.1	The James Construction	31
7.2	The Infinite Symmetric Product	31
8	Products between Homotopy Groups	33
8.1	The Samelson Product	33
8.2	The Whitehead Product	33
8.3	The Hilton Milnor Theorem	34
III	Topics in Algebraic Topology	35
9	The Eilenberg-Steenrod Axioms	36

9.1	Generalized (Co)Homology Theories for Spaces	36
9.2	Reduced (Co)Homology Theory for CW Complexes	38
9.3	Brown Representability Theorem	40
10	Stable Phenomena	41
10.1	The Stable Homotopy Groups	41
10.2	Spanier Whitehead Duality	42
11	Spectral Sequences in Algebraic Topology	43
11.1	Spectral Sequences in Topology	43
12	Bonus?	45
12.1	Postnikov Towers	45
12.2	Reading Cell Structures from Homology Groups	45

Part I

Some Point Set Topology

1 The Compact Open Topology

1.1 The Mapping Space

Definition 1.1.1 (Compact Open Functions)

Let X, Y be spaces. Let $K \subseteq X$ be compact. Let $U \subseteq Y$ be open. Define

$$V(K, U) = \{f : X \rightarrow Y \mid f \text{ is a map and } f(K) \subseteq U\}$$

Definition 1.1.2 (The Mapping Space)

Let X, Y be spaces. Define the mapping space of X and Y to be the set

$$\text{Map}(X, Y) = \{f : X \rightarrow Y \mid f \text{ is continuous}\}$$

together with the topology generated by the subbase

$$\{V(K, U) \mid K \subseteq X \text{ is compact and } U \subseteq Y \text{ is open}\}$$

Recall that the exponential law of sets says that

$$\text{Hom}_{\text{Set}}(X \times Y, Z) \cong \text{Hom}_{\text{Set}}(X, \text{Hom}_{\text{Set}}(Y, Z))$$

When we upgrade $\text{Hom}_{\text{Set}}(Y, Z)$ to the mapping space, the exponential law no longer holds in general, unless Y satisfies additional assumptions.

Proposition 1.1.3

Let X, Y, Z be spaces. If Y is locally compact and Hausdorff, then the exponential law induces a natural bijection of sets

$$\text{Hom}_{\text{Top}}(X \times Y, Z) \cong \text{Hom}_{\text{Top}}(X, \text{Map}(Y, Z))$$

Proposition 1.1.4

Let X, Y, Z be spaces. If Y is locally compact and Hausdorff, then the above bijection induces a homeomorphism

$$\text{Map}(X \times Y, Z) \cong \text{Map}(X, \text{Map}(Y, Z))$$

Definition 1.1.5 (The Pointed Mapping Space)

Let X, Y be pointed spaces. Define the mapping space of X and Y to be the set

$$\text{Map}_*(X, Y) = \{f : X \rightarrow Y \mid f \text{ is a pointed map}\}$$

together the topology given by ???

1.2 The Smash Product and the Pointed Mapping Space

We would like to have a theorem similar to the exponential law in pointed spaces. The correct object for the adjunction is the smash product instead of the usual Cartesian product.

Definition 1.2.1 (The Smash Product)

Let (X, x_0) and (Y, y_0) be pointed topological spaces. Define the smash product of the two pointed spaces to be the space

$$X \wedge Y = \frac{X \times Y}{X \vee Y}$$

together with the basepoint (x_0, y_0) .

Proposition 1.2.2

Let X, Y, Z be pointed spaces. If Y is locally compact and Hausdorff, then the exponential law

induces a natural bijection of sets

$$\mathrm{Hom}_{\mathbf{Top}_*}(X \wedge Y, Z) \cong \mathrm{Hom}_{\mathbf{Top}_*}(X, \mathrm{Map}_*(Y, Z))$$

Proposition 1.2.3

Let X, Y, Z be pointed spaces. If Y is locally compact and Hausdorff, then the above bijection induces a homeomorphism

$$\mathrm{Map}_*(X \wedge Y, Z) \cong \mathrm{Map}_*(X, \mathrm{Map}_*(Y, Z))$$

2 Suspension and Loopspaces

2.1 Suspensions and Reduced Suspensions

Definition 2.1.1 (Reduced Suspension) Let (X, x_0) be a pointed space. Define the reduced suspension of X to be the space

$$\Sigma X = \frac{X \times I}{(X \times \{0\} \cup X \times \{1\} \cup \{x_0\} \times I)}$$

together with the base point as the equivalence class in the quotient.

By definition of quotients, the reduced suspension defines a canonical continuous map sending a space X to its reduced suspension ΣX .

Proposition 2.1.2 The reduced suspension Σ define functors on $\mathbf{Top}_*$ and $\mathbf{hTop}_*$ as follows.

- Σ sends a pointed space (X, x_0) to

$$(\Sigma X, (x_0, 0))$$

- For the non homotopy category, Σ sends a map $f : X \rightarrow Y$ to

$$\Sigma f : \Sigma X \rightarrow \Sigma Y$$

defined by $\Sigma f([x, t]) = [f(x), t]$

- For the homotopy category, Σ sends a homotopy class of maps $[X, Y]$ to

$$[\Sigma X, \Sigma Y]$$

respectively given by the same formula as above

Example 2.1.3 Let $n \in \mathbb{N}$. Then there is a homeomorphism $\Sigma S^n \cong S^{n+1}$.

Proposition 2.1.4

Let X be a space. Let R be a ring. Then the following are true.

- $\tilde{H}_{k+1}(\Sigma X) \cong \tilde{H}_k(X)$ for all $k \geq 0$.
- $\tilde{H}^{k+1}(\Sigma X; R) \cong \tilde{H}^k(X; R)$ for all $k \geq 0$.
- All cup products in $H^*(X; R)$ are 0.

Definition 2.1.5 ((Reduced) Cone) Let (X, x_0) be a pointed space. Define the reduced cone of the space to be

$$CX = \frac{X \times I}{X \times \{1\} \cup \{x_0\} \times I}$$

together with the base point $[(x_0, 1)] = X \times \{1\} \cup \{x_0\} \times I$.

Proposition 2.1.6 The reduced cone defines a functor on

$$C : \mathbf{Top}_* \rightarrow \mathbf{Top}_*$$

as follows.

- C sends a pointed space (X, x_0) to $(CX, [(x_0, 1)])$
- C sends a map $f : X \rightarrow Y$ to

$$Cf : CX \rightarrow CY$$

given by the induced map $f \times \text{id}_I$ by taking quotients.

Definition 2.1.7 (Unreduced Cone) Let X be a space. Define the unreduced cone of X to be

$$CX = \frac{X \times I}{X \times \{1\}}$$

Definition 2.1.8 (Unreduced Suspensions) Let X be a space. Define the unreduced suspension of X to be the space

$$SX = \frac{X \times I}{X \times \{0\}, X \times \{1\}}$$

Proposition 2.1.9 Let X be a space. Then the following diagram is a pushout:

$$\begin{array}{ccc} X \times \{0\} & \hookrightarrow & CX \\ \downarrow & & \downarrow \\ CX & \longrightarrow & SX \end{array}$$

Proposition 2.1.10 Let X be a space. The the unreduced suspension map induces natural isomorphisms

$$H_{k+1}(SX) \cong H_k(SX)$$

for $k \geq 1$. Moreover, the following are true.

- There is an isomorphism $H_0(X) \cong H_1(SX) \oplus \mathbb{Z}$
- $H_0(SX) \cong \mathbb{Z}$.

2.2 Loopspaces and Free Loopspaces

Definition 2.2.1 (Loop Spaces)

Let X be a pointed space. Define the loop space of X to be

$$\Omega X = \text{Map}_*(S^1, X)$$

Lemma 2.2.2

Let G be an abelian group and let $n \in \mathbb{N}$. Then there is a homeomorphism

$$\Omega K(G, n) \cong K(G, n-1)$$

Definition 2.2.3 (Free Loopspace)

Let X be a space. Define the free loop space of X to be

$$LX = \text{Map}(S^1, X)$$

2.3 The Suspension-Loopspace Adjunction

Theorem 2.3.1 (The Suspension-Loopspace Adjunction)

There is an adjunction $\Sigma : \mathbf{Top}_* \xrightarrow{-1} \mathbf{Top}_* : \Omega$. Explicitly, there is a natural bijection of sets

$$\text{Hom}_{\mathbf{Top}_*}(\Sigma X, Y) \cong \text{Hom}_{\mathbf{Top}_*}(X, \Omega Y)$$

for any spaces $X, Y \in \mathbf{Top}_*$.

Proposition 2.3.2

Let X, Y be pointed spaces. Then the above bijection induces a homeomorphism

$$\text{Map}_*(\Sigma X, Y) \cong \text{Map}_*(X, \Omega Y)$$

Corollary 2.3.3

The above adjunction induces an adjunction $\Sigma : \mathbf{hTop}_* \xrightleftharpoons{\perp} \mathbf{hTop}_* : \Omega$. Explicitly, there is a natural bijection of sets

$$[\Sigma X, Y]_* \cong [X, \Omega Y]_*$$

for any spaces $X, Y \in \mathbf{Top}_*$.

Part II**The Higher Homotopy Groups and Their Properties**

3 The Higher Homotopy Groups

The journey of Algebraic Topology began with the fundamental group, where we assigned a group to every space functorially. The notion of fundamental group heavily involves the notion of homotopy and therefore is heavily related to the notion of homotopy. However, one realizes that even with Seifert-van Kampen theorem and the theory of covering spaces, it is not easy to compute the fundamental group of a space. This is partly, but not wholly due to the fundamental group is in general not abelian. If we instead work in an abelian setting, one is able to distinguish two non-isomorphic groups simply by analysing the torsion subgroups. Therefore we refine the concept of the fundamental group and procured the notion of homology and cohomology. Both functorial invariants now produce graded abelian groups for each space, one for each dimension $n \in \mathbb{N}$. In the case of cohomology, there is a canonical ring structure on cohomology that interacts with the topology of the underlying space.

Now we turn to the final main invariant of topological spaces. The homotopy groups $\pi_n(X, x_0)$ serves as both a generalization of the fundamental group $\pi_1(X, x_0)$ in higher dimensions and a homotopic analogue to homology via the Hurewicz homomorphism

$$h : \pi_n(X) \rightarrow H_n(X)$$

It is a strong invariant that is closely related to the notion of homotopy, all the while having mostly abelian groups as its output. The trade off is that the homotopy groups are very hard to compute. Such trade off has led to the blossoming of Algebraic Topology in its fullest. For instance, stable homotopy theory stems from a crucial fact called the Freudenthal suspension theorem, which states that such a sequence

$$\pi_n(X) \rightarrow \pi_{n+1}(\Sigma X) \rightarrow \cdots$$

eventually stabilizes for large enough n .

In this chapter we will closely study the n th homotopy groups such as its properties and develop tools to compute them.

3.1 The n th Homotopy Groups

We begin not with the definition of the homotopy groups, but rather a slight generalization of pointed spaces and maps between them.

Definition 3.1.1 (Pairs of Space) Let X be a topological space. A pair of space is a pair (X, A) where $A \subseteq X$ is a subspace of X . A map of pairs $f : (X, A) \rightarrow (Y, B)$ is a continuous map $f : X \rightarrow Y$ such that $f(A) \subseteq B$.

Definition 3.1.2 (Homotopy between Maps of Pairs) Let $f, g : (X, A) \rightarrow (Y, B)$ be maps of pairs. A homotopy between f and g is a homotopy $H : X \times [0, 1] \rightarrow Y$ such that $H(A \times [0, 1]) \subseteq B$.

Definition 3.1.3 (The n th Homotopy Groups) Let (X, x_0) be a pointed space. Define the n th homotopy group $\pi_n(X, x_0)$ to be

$$\pi_n(X, x_0) = \frac{\left\{ \gamma : (I^n, \partial I^n) \rightarrow (X, \{x_0\}) \mid \gamma \text{ is continuous} \right\}}{\simeq}$$

where we say that $f \simeq g$ if there exists a homotopy between f and g .

Notice that the definition coincides with that of the fundamental group when $n = 1$, and hence π_n is indeed a generalization.

Now we have defined the set $\pi_n(X, x_0)$ for a pointed space to have the word group in its name. We will also need to procure a canonical group structure on the set $\pi_n(X, x_0)$. This will be similar with that of the fundamental group.

Definition 3.1.4 (Concatenation) Let $n \geq 1$. Let (X, x_0) be a pointed space. Let $f, g : (I^n, \partial I^n) \rightarrow (X, x_0)$ be maps. Define the composition of f and g by the formula

$$(f \cdot g)(t_1, \dots, t_n) = \begin{cases} f(2t_1, t_2, \dots, t_n) & \text{if } 0 \leq t_1 \leq \frac{1}{2} \\ g(2t_1 - 1, t_2, \dots, t_n) & \text{if } \frac{1}{2} \leq t_1 \leq 1 \end{cases}$$

for $f, g \in \pi_n(X, x_0)$.

Notice that concatenation is really just the same concatenation between elements of the fundamental group but instead with more coordinates. The group structure on $\pi_n(X, x_0)$ uses concatenation and such a proof also uses the same homotopies as in Algebraic Topology 1, but with more coordinates.

Theorem 3.1.5 Let (X, x_0) be a pointed space and $n \geq 1$. The operation $\cdot$ on the equivalence classes in $\pi_n(X, x_0)$ is well defined and endows it with the structure of a group.

Proof We first show that the operation is well defined on $\pi_n(X, x_0)$. Suppose that $f_1 \stackrel{\partial}{\simeq} g_1 : (I^n, \partial I^n) \rightarrow (X, x_0)$ via the homotopy H_1 and $f_2 \stackrel{\partial}{\simeq} g_2 : (I^n, \partial I^n) \rightarrow (X, x_0)$ via the homotopy H_2 . Consider the map $H : I^n \times [0, 1] \rightarrow X$ defined by

$$H(x_1, \dots, x_n, t) = \begin{cases} H_1(2x_1, \dots, x_n, t) & \text{if } 0 \leq x_1 \leq \frac{1}{2} \\ H_2(2x_1 - 1, \dots, x_n, t) & \text{if } \frac{1}{2} \leq x_1 \leq 1 \end{cases}$$

Now when $t = 0$, we have that $H(x_1, \dots, x_n, 0) = f_1 \cdot f_2$. When $t = 1$, we have that $H(x_1, \dots, x_n, 1) = g_1 \cdot g_2$. Now notice that by definition of H_1 and H_2 , if one of $x_1, \dots, x_n$ is equal to 0 or 1, then H_1 and H_2 is constant and maps to x_0 . This means that H also has such property and hence H is a homotopy $(I, \partial I^n)$ to (X, x_0) .

We now have an appropriate binary operation on $\pi_n(X, x_0)$. It is clearly associative since the composition of maps are associativity and one can re-parametrize homotopies with different traversal speeds. I claim that the constant map $e_{x_0} : (I, \partial I^n) \rightarrow (X, x_0)$ defined by $e_{x_0}(x) = x_0$ is the identity. Let $f : (I^n, \partial I^n) \rightarrow (X, x_0)$ be arbitrary. Define the homotopy from $e_{x_0} \cdot f$ to f by

$$H(x_1, \dots, x_n, t) = \begin{cases} e_{x_0}(x_1, \dots, x_n) = x_0 & \text{if } 0 \leq x_1 \leq \frac{1-t}{2} \\ f\left(\frac{2s+t-1}{t+1}\right) & \text{if } \frac{1-t}{2} \leq x_1 \leq 1 \end{cases}$$

A similar homotopy proves that $f \cdot e_{x_0} \simeq f$. For the inverse, I claim that $\bar{f} : (I^n, \partial I^n) \rightarrow (X, x_0)$ defined by $\bar{f}(1 - x_1, \dots, x_n)$ is the inverse of f . Indeed, define a homotopy from $f \cdot \bar{f}$ to e_{x_0} by

$$H(x_1, \dots, x_n, t) = \begin{cases} e_{x_0}(x_1, \dots, x_n) = x_0 & \text{if } 0 \leq x_1 \leq \frac{t}{2} \text{ or } \frac{1-t}{2} \leq x_1 \leq 1 \\ f(2x_1 - t, x_2, \dots, x_n) & \text{if } \frac{t}{2} \leq x_1 \leq \frac{1}{2} \\ \bar{f}(2s + t - 1) & \text{if } \frac{1}{2} \leq x_1 \leq \frac{1-t}{2} \end{cases}$$

■

However, what makes each $\pi_n(X, x_0)$ for $n \geq 2$ different from the fundamental group $\pi_1(X, x_0)$ is the abelian group structure on $\pi_n(X, x_0)$.

Theorem 3.1.6 Let (X, x_0) be a pointed space. Then the n th homotopy group

$$\pi_n(X, x_0)$$

together with concatenation is abelian.

Proof Define a new operation $\star : \pi_n(X, x_0) \times \pi_n(X, x_0) \rightarrow \pi_n(X, x_0)$ by

$$[f] \star [g] = \begin{cases} f(t_1, 2t_2, \dots, t_n) & \text{if } 0 \leq t_1 \leq \frac{1}{2} \\ g(t_1, 2t_2 - 1, \dots, t_n) & \text{if } \frac{1}{2} \leq t_1 \leq 1 \end{cases}$$

Such an operation clearly also defines an abelian group structure on $\pi_n(X, x_0)$ using the same argument. Now I want to prove that

$$([f] \star [g]) \star ([h] \star [k]) = ([f] \star [h]) \star ([g] \star [k])$$

This is true because

$$([f] \star [g]) \star ([h] \star [k]) = \begin{cases} f(2x_1, 2x_2, x_3, \dots, x_n) & \text{if } 0 \leq x_1, x_2 \leq \frac{1}{2} \\ g(2x_1, 2x_2 - 1, x_3, \dots, x_n) & \text{if } 0 \leq x_1 \leq \frac{1}{2} \text{ and } \frac{1}{2} \leq x_2 \leq 1 \\ h(2x_1 - 1, 2x_2, x_3, \dots, x_n) & \text{if } \frac{1}{2} \leq x_1 \leq 1 \text{ and } 0 \leq x_2 \leq \frac{1}{2} \\ k(2x_1, 2x_2 - 1, x_3, \dots, x_n) & \text{if } \frac{1}{2} \leq x_1, x_2 \leq 1 \end{cases}$$

and

$$([f] \star [h]) \star ([g] \star [k]) = \begin{cases} f(2x_1, 2x_2, x_3, \dots, x_n) & \text{if } 0 \leq x_1, x_2 \leq \frac{1}{2} \\ h(2x_1 - 1, 2x_2, x_3, \dots, x_n) & \text{if } \frac{1}{2} \leq x_1 \leq 1 \text{ and } 0 \leq x_2 \leq \frac{1}{2} \\ g(2x_1, 2x_2 - 1, x_3, \dots, x_n) & \text{if } 0 \leq x_1 \leq \frac{1}{2} \text{ and } \frac{1}{2} \leq x_2 \leq 1 \\ k(2x_1, 2x_2 - 1, x_3, \dots, x_n) & \text{if } \frac{1}{2} \leq x_1, x_2 \leq 1 \end{cases}$$

which are entirely the same. Now I claim that $\star = \star$. It is clear that both binary operations have the same identity element e_{x_0} . Now we have that

$$f \star g = (f \star 1) \star (1 \star g) = (f \star 1) \star (1 \star g) = f \star g$$

Finally, I claim that $\star$ is commutative. We have that

$$f \star g = (1 \star f) \star (g \star 1) = (1 \star g) \star (f \star 1) = g \star f = g \star f$$

Thus we conclude. ■

The above technique is actually called the Eckmann-Hilton argument. In particular, it shows that concatenation of paths need not be defined via the first coordinate. Any choice of coordinate to perform concatenation will result in the same group structure.

Example 3.1.7

The homotopy groups of the sphere is given by

$$\pi_n(S^1) \cong \begin{cases} \mathbb{Z} & \text{if } n = 1 \\ 0 & \text{otherwise} \end{cases}$$

Proof

We have seen that $\pi_0(S^1) = 0$ and $\pi_1(S^1) \cong \mathbb{Z}$.

When $n \geq 2$, suppose that $[f] \in \pi_n(S^1)$. Then we have a map $f : S^k \rightarrow S^1$. By the lifting criterion in Algebraic Topology 1, we may lift the map to the covering space $p : \mathbb{R} \rightarrow S^1$ since $0 = f_*(\pi_1(S^k)) \leq p_*(\pi_1(\mathbb{R})) = 0$. Hence f factors through a contractible space, and so is null homotopic. Hence $[f] = 0$. ■

Lemma 3.1.8

For any $n \in \mathbb{N}^+$, the two spaces $(I^n, \partial I^n)$ and (S^n, s_0) are homotopy equivalent.

Corollary 3.1.9

Let (X, x_0) be a pointed space. Then we have

$$\pi_n(X, x_0) = [(S^n, 1), (X, x_0)]_*$$

Therefore an alternate viewpoint of the homotopy groups is instead the collection of maps from the pointed n -sphere to the space X quotient homotopy. Indeed an n -dimensional sphere has an n -dimensional hole enclosed by the sphere itself. Therefore in order to detect n -dimensional holes in a space, we are permitted to try and fit n -spheres into the space.

Spheres are also advantageous for the definition of π_n because spheres only has an n -dimensional hole and no other holes in any dimension. Therefore we are capturing the minimal amount of information on n -dimensional holes without producing excess data.

The group structure of π_n is also easy to understand in terms of spheres.

Lemma 3.1.10

Let (X, x_0) be a pointed space. Let $f, g : S^n \rightarrow X$ be pointed maps. Then $[f] \cdot [g] \in \pi_n(X, x_0)$ is represented by the map

$$S^n \xrightarrow{\text{proj.}} S^n \vee S^n \xrightarrow{f \vee g} X$$

where the projection map is given by $S^n \vee S^n \cong \frac{S^n}{S^{n-1}}$ where $S^{n-1} \hookrightarrow S^n$ is the inclusion along the equator.

3.2 Functorial Properties of Homotopy

The homotopy groups also satisfy functorial properties similar to the fundamental group and the (co)homology groups.

Theorem 3.2.1 (Functoriality) Let (X, x_0) and (Y, y_0) be pointed spaces and let $f : (X, x_0) \rightarrow (Y, y_0)$ be a pointed map. Then the induced map

$$\pi_n(f) : \pi_n(X, x_0) \rightarrow \pi_n(Y, y_0)$$

defined by $[\gamma] \mapsto [f \circ \gamma]$ is a group homomorphism. Moreover, it satisfies the following functorial properties.

- If $g : (Y, y_0) \rightarrow (Z, z_0)$ is a pointed map then

$$\pi_n(g \circ f) = \pi_n(g) \circ \pi_n(f)$$

- If $\text{id}_{(X, x_0)} : (X, x_0) \rightarrow (X, x_0)$ is the identity map then

$$\pi_n(\text{id}_{(X, x_0)}) = \text{id}_{\pi_n(X, x_0)}$$

Proof Firstly, let us show that it is a group homomorphism. Let $\gamma_1, \gamma_2 \in \pi_n(X, x_0)$. We have that

$$\pi_n(f)([\gamma_1] \cdot [\gamma_2]) = [f \circ (\gamma_1 \cdot \gamma_2)] = [f \circ \gamma_1 \cdot f \circ \gamma_2] = \pi_n(f)([\gamma_1]) \cdot \pi_n(f)([\gamma_2])$$

where the second equality is true because homotopies are preserved under function composition. It remains to show associativity and unitality.

- Associativity: We have that

$$\pi_n(g \circ f)([\gamma]) = [g \circ f \circ \gamma] = \pi_n(g)([f \circ \gamma]) = (\pi_n(g) \circ \pi_n(f))([\gamma])$$

- Unitality: We have that

$$\pi_n(\text{id}_{(X, x_0)})([\gamma]) = [\text{id}_{(X, x_0)} \circ \gamma] = [\gamma] = \text{id}_{\pi_n(X, x_0)}([\gamma])$$

And so we conclude. ■

Similar to all other functorial properties we have seen throughout algebraic topology, a homeomorphism of spaces give an isomorphism on homotopy groups. Now that we know about category theory, we see that such a result does not depend on the definition of the homotopy groups or the (co)homology groups, but is in fact due to the functorial properties of each invariant.

Similar to (co)homology and the fundamental group, the homotopy groups are defined via a quotient with homotopy. Therefore we expect the homotopy groups to not be able to distinguish between homotopy equivalent spaces but not homeomorphic spaces.

Theorem 3.2.2 (Homotopy Equivalence) Let $(X, x_0), (Y, y_0)$ be pointed spaces and $f, g : (X, x_0) \rightarrow (Y, y_0)$ be pointed maps. If f and g are homotopic, then the induced maps

$$\pi_n(f) = \pi_n(g) : \pi_n(X, x_0) \rightarrow \pi_n(Y, y_0)$$

are equal. Moreover, if f is a homotopy equivalence, then $\pi_n(f)$ is an isomorphism.

Proof Let $[\gamma] \in \pi_n(X, x_0)$. Suppose that f and g are homotopic via $F : X \times I \rightarrow Y$. now define

$$H(x_1, \dots, x_n, t) = F(\gamma(x_1, \dots, x_n), t)$$

Then it is clear that $H(x_1, \dots, x_n, 0) = f \circ \gamma$ and $H(x_1, \dots, x_n, 1) = g \circ \gamma$. Thus $[f \circ \gamma] = [g \circ \gamma]$ and so we conclude that $\pi_n(f)([\gamma]) = \pi_n(g)([\gamma])$.

If f is a homotopy equivalence, then there exists $g : (Y, y_0) \rightarrow (X, x_0)$ such that $g \circ f \simeq \text{id}_{(X, x_0)}$ and $f \circ g \simeq \text{id}_{(Y, y_0)}$. By functoriality and homotopy equivalence, we have that

$$\pi_n(g) \circ \pi_n(f) = \text{id}_{\pi_n(X, x_0)} \quad \text{and} \quad \pi_n(f) \circ \pi_n(g) = \text{id}_{\pi_n(Y, y_0)}$$

and so we conclude. ■

Similar to the fundamental group, changing the base point via a path induces isomorphisms on homotopy groups with the same space but different base point.

Theorem 3.2.3 Let (X, x_0) and (X, x_1) be pointed spaces with the same base space. Let $u : I \rightarrow X$ be a path from x_0 to x_1 . Define the induced map

$$u_{\#} : \pi_n(X, x_0) \rightarrow \pi_n(X, x_1)$$

as follows. For $[\gamma] \in \pi_n(X, x_0)$ define $u_{\#}([\gamma])$ by first shrinking the domain of γ to a smaller concentric cube in I^n . Then inserting the path γ on each radical segment of the shell between the smaller cube and ∂I^n .

The construction of $u_{\#}$ is a group isomorphism. Moreover, it satisfies the following universal properties.

- If $v : I \rightarrow X$ is a path from x_1 to x_2 and $u \cdot v$ is the concatenation of these paths, then

$$(u \cdot v)_{\#} = u_{\#} \circ v_{\#}$$

- If c_{x_0} is the constant path from x_0 to x_0 then $(c_{x_0})_{\#}$ is the identity

Proposition 3.2.4 Let X_i for $i \in I$ be a family of path connected spaces. Then there are isomorphisms

$$\pi_n \left(\prod_{i \in I} X_i \right) \cong \prod_{i \in I} \pi_n(X_i)$$

Proposition 3.2.5 Let (X, x_0) and (X, x_1) be pointed spaces with the same base space. Let $u, v : I \rightarrow X$ be paths from x_0 to x_1 . If u and v are homotopic relative to end points then the induced

maps

$$u_{\#} = v_{\#} : \pi_n(X, x_0) \rightarrow \pi_n(X, x_1)$$

are equal.

This shows that if X is path connected, then $\pi_n(X, x_0)$ no longer depends on the choice of base point. Although there are no canonical isomorphisms between $\pi_n(X, x_0)$ and $\pi_n(X, x_1)$, we still forget about the base point in this case and write the homotopy groups as $\pi_n(X)$.

3.3 The Action of the Fundamental Group

Proposition 3.3.1

Let (X, x_0) be a pointed space. Let $n \geq 1$. Then the map $\pi_1(X, x_0) \times \pi_n(X, x_0) \rightarrow \pi_n(X, x_0)$ defined by

$$([u], [f]) \mapsto u_{\#}(f)$$

is a group action on $\pi_n(X, x_0)$.

When $n = 1$, the above group action is just left multiplication in $\pi_1(X, x_0)$.

3.4 Relative Homotopy Groups

Definition 3.4.1 (Triplets of Spaces) Let X be a topological space. A pointed pair of space is a triple (X, A_1, A_2) where $A_2 \subseteq A_1 \subseteq X$ are subspaces of X . A map between triplets of spaces $f : (X, A_1, A_2) \rightarrow (Y, B_1, B_2)$ is a map $f : X \rightarrow Y$ such that $f(A_1) \subseteq B_1$ and $f(A_2) \subseteq B_2$. If $A_2 = \{x_0\}$ is a single point we say that (X, A, x_0) is a pointed pair of spaces.

Definition 3.4.2 (Homotopy between Maps of Triplets) Let $f, g : (X, A_1, A_2) \rightarrow (Y, B_1, B_2)$ be maps triplets of spaces. A homotopy between f and g is a homotopy between $f : X \rightarrow Y$ and $g : X \rightarrow Y$, namely $H : X \times [0, 1] \rightarrow Y$ such that $H(A_1 \times [0, 1]) \subseteq B_1$ and $H(A_2 \times [0, 1]) \subseteq B_2$.

Definition 3.4.3 (The n th Relative Homotopy Groups) Let (X, A, x_0) be a pointed pair of space. Let $n \geq 2$. Regard I^{n-1} sitting inside I^n by $I^{n-1} = \{(x_1, \dots, x_n) \in I^n \mid x_n = 0\}$ and let $J^{n-1} = \partial I^n \setminus I^{n-1}$. Define the relative homotopy groups of the triple by

$$\pi_n(X, A, x_0) = \frac{\left\{ \gamma : (I^n, \partial I^n, J^{n-1}) \rightarrow (X, A, x_0) \mid \gamma \text{ is continuous} \right\}}{\simeq}$$

where we say that $f \simeq g$ if there exists a homotopy between f and g .

It is easy to see that $\pi_n(X, x_0, x_0) = \pi_n(X, x_0)$ so that homotopy groups are a special case of the relative homotopy groups.

Lemma 3.4.4 For any $n \in \mathbb{N}$, the two triplets $(I^n, \partial I^n, J^{n-1})$ and (D^n, S^{n-1}, s_0) are homotopy equivalent.

Theorem 3.4.5 Let (X, A, x_0) be a pointed pair of space. The composition law on definition 1.1.4 defines a group structure on $\pi_n(X, A, x_0)$ for $n \geq 2$. Moreover, $\pi_n(X, A, x_0)$ is abelian for $n \geq 3$.

3.5 Induced Maps of Relative Homotopy Groups

Theorem 3.5.1 Let (X, A, x_0) and (Y, B, y_0) be pointed pairs of spaces and $f : (X, A, x_0) \rightarrow (Y, B, y_0)$ a map. Then f induces a map on the relative homotopy groups

$$f_* : \pi_n(X, A, x_0) \rightarrow \pi_n(Y, B, y_0)$$

for $n \geq 2$ satisfying the following functorial properties:

- f_* is a group homomorphism
- If $g : (Y, B, y_0) \rightarrow (Z, C, z_0)$ is a map, then

$$(g \circ f)_* = g_* \circ f_*$$

- If $\text{id}_{(X, A, x_0)}$ is the identity map on (X, A, x_0) , then

$$(\text{id}_{(X, A, x_0)})_* = \text{id}_{\pi_n(X, A, x_0)}$$

Theorem 3.5.2 Let $(X, A, x_0), (Y, B, y_0)$ be pointed pairs of spaces and $f, g : (X, A, x_0) \rightarrow (Y, B, y_0)$ be pointed maps. If f and g are homotopic, then the induced maps

$$f_* = g_* : \pi_n(X, A, x_0) \rightarrow \pi_n(Y, B, y_0)$$

are equal. Moreover, if f is a homotopy equivalence, then f_* is an isomorphism.

TBA: change of base point isomorphisms.

3.6 Long Exact Sequence in Homotopy Groups

Lemma 3.6.1 (Compression Criterion) Let (X, A, x_0) be a pair of spaces with basepoint. Let $f : (D^n, S^{n-1}, *) \rightarrow (X, A, x_0)$ be a map. Then $[f] = [e_{x_0}] \in \pi_n(X, A, x_0)$ if and only if

$$(f : D^n \rightarrow X) \stackrel{S^{n-1}}{\simeq} (g : D^n \rightarrow X)$$

where g is any map such that $g(X) \subseteq A$.

Proof Suppose that the second criterion is satisfied. Then it clearly shows that $[f] = [g] \in \pi_n(X, A, x_0)$. Let $r : D^n \times I \rightarrow D^n$ be a deformation retract from D^n to $*$ in $S^{n-1} \subset D^n$. Consider the map $g \circ r : D^n \times I \rightarrow X$. When $t = 0$, this is the map g . When $t = 1$, $g \circ r$ factors through $*$ and so becomes a map $*$ $\rightarrow X$. In other words, it is the constant map e_{x_0} . Moreover, it $g \circ r$ has image in A and so in particular it sends S^{n-1} to A . Thus $g \circ r$ is a homotopy between e_{x_0} and g . We conclude that $[f] = [g] = [e_{x_0}]$.

Now suppose that $[f] = [e_{x_0}] \in \pi_n(X, A, x_0)$ is given by the homotopy $H : D^n \times I \rightarrow X$. This means that $H(D^n \times \{1\}) \subseteq \{x_0\} \subset A$ and $H(S^{n-1} \times I) \subset A$. Now $D^n \times I$ deformation retracts to the cup $D^n \times \{1\} \cup S^{n-1} \times I$ by radial projection from the center point of $D^n \times \{0\}$. Thus H can be converted into a map from $D^n \times \{1\} \cup S^{n-1} \times I$ to X . Then H is now a homotopy from f to a map $H(-, 1) : D^n \rightarrow X$ which has image in A , relative to S^{n-1} . Thus we conclude. ■

Theorem 3.6.2 Let X be a space and A, B be subspaces of X such that $B \subseteq A \subseteq X$. Let $x_0 \in B$. Then there is a long exact sequence in relative homotopy groups:

$$\cdots \longrightarrow \pi_n(A, B, x_0) \xrightarrow{i_*} \pi_n(X, B, x_0) \xrightarrow{j_*} \pi_n(X, A, x_0) \xrightarrow{\partial_n} \pi_{n-1}(A, B, x_0) \longrightarrow \cdots \longrightarrow \pi_1(X, A, x_0)$$

where $i : (A, B, x_0) \rightarrow (X, B, x_0)$ and $j : (X, B, x_0) \rightarrow (X, A, x_0)$ are the inclusions and $\partial : \pi_n(X, A, x_0) \rightarrow \pi_{n-1}(A, B, x_0)$ is given by $[\gamma] \mapsto [\gamma|_{I^{n-1}}]$

Proof ■

TBA: Naturality of the sequence.

Theorem 3.6.3 Let (X, A, x_0) be a pointed pair of spaces. The relative homotopy groups and (absolute) homotopy groups of (X, A, x_0) fit into a long exact sequence

$$\cdots \longrightarrow \pi_{n+1}(X, A, x_0) \xrightarrow{\partial_{n+1}} \pi_n(A, x_0) \xrightarrow{i_*} \pi_n(X, x_0) \xrightarrow{j_*} \pi_n(X, A, x_0) \xrightarrow{\partial_n} \pi_{n-1}(A, x_0) \longrightarrow \cdots \longrightarrow \pi_0(X, x_0) \longrightarrow 0$$

where ∂_n is defined by $[f] \mapsto [f|_{I^{n-1}}]$ and i_* and j_* are induced by inclusions.

Note that even though at the end of the sequence group structures are not defined, exactness still makes sense: kernels in this case consists of elements that map to the homotopy class of the constant map.

3.7 Homotopical Connectivity

Definition 3.7.1 (n-Connected Space)

Let X be a space. We say that it is n -connected if

$$\pi_k(X, x_0) = 0$$

for $0 \leq k \leq n$ and some $x_0 \in X$.

Note that $\pi_0(X, x_0)$ implies that X is path connected. Hence the notion of n -connectedness does not depend on the base point by the change of base point isomorphism. In particular, $\pi_k(X, x_0) = 0$ for $0 \leq k \leq n$ and some $x_0 \in X$ if and only if $\pi_k(X, x_0) = 0$ for $0 \leq k \leq n$ for all $x_0 \in X$. (Hatcher)

Definition 3.7.2 (n-Connected Pair of Spaces)

Let (X, A) be a pair of space. We say that it is n -connected if the following are true.

- $\pi_k(X, A, x_0) = 0$ for $0 < k \leq n$ and all $x_0 \in A$.
- $\pi_0(\iota) : \pi_0(A) \rightarrow \pi_0(X)$ is surjective.

TBA: conditions in P.346 of Hatcher

Definition 3.7.3 (Weakly Contractible)

Let X be a space. We say that X is weakly contractible if

$$\pi_n(X) = 0$$

for all $n \geq 0$.

Definition 3.7.4 (Weak Homotopy Equivalence) We say that a map $f : X \rightarrow Y$ is a weak homotopy equivalence if it induces isomorphisms on all homotopy groups π_n on any choice of base point.

TBA: compression lemma in Hatcher

Proposition 3.7.5

Let X, Y be spaces and let $f : X \rightarrow Y$ be a weak homotopy equivalence. Then f induces bijections

$$[Z, X] \cong [Z, Y] \quad \text{and} \quad [Z, X]_* \cong [Z, Y]_*$$

for all CW-complexes Z .

4 Whitehead's Theorem and CW Approximations

4.1 Whitehead's Theorem

Theorem 4.1.1 (Whitehead's Theorem) If X and Y are CW-complexes and $f : X \rightarrow Y$ is a weak homotopy equivalence, then f is a homotopy equivalence.

TBA: extension lemma in Hatcher.

Corollary 4.1.2 If X and Y are CW-complexes with $\pi_1(X) = \pi_1(Y) = 0$ and $f : X \rightarrow Y$ induces isomorphisms on homology groups H_n for all n , then f is a homotopy equivalence.

4.2 Cellular Approximations

Definition 4.2.1 (Cellular Approximations) Let X and Y be CW-complexes. We say that $f : X \rightarrow Y$ has a cellular approximations if f is homotopic to a cellular map $f' : X \rightarrow Y$.

To this end we need to revisit the notion of polyhedra.

Definition 4.2.2 (Convex Polyhedra) Let $n \in \mathbb{N}$. A convex polyhedra is a subset S of $\mathbb{R}^n$ of the form

$$S = \left\{ x = (x_1, \dots, x_n) \in \mathbb{R}^n \mid \sum_{k=1}^n a_{k,1}x_k \leq b_1, \dots, \sum_{k=1}^n a_{k,s}x_k \leq b_s \right\}$$

for some $a_{1,1}, \dots, a_{n,s}, b_1, \dots, b_s \in \mathbb{R}$.

Lemma 4.2.3 Let X be a CW complex. Let $Y \subseteq X$ be a CW subcomplex. Let Z be obtained by attaching a cell e^k to Y . Let $f : I^n \rightarrow Z$ be a map. Then there exists a map $g : I^n \rightarrow X$ such that

$$f \stackrel{f^{-1}(Y)}{\simeq} g$$

and g is such that the following is true. There exists a simplex $\Delta^k \subset e^k$ and polyhedra $P_1, \dots, P_d$ such that g is the (possibly empty) union

$$g^{-1}(\Delta^k) = \bigcup_{k=1}^n$$

and $g|_{P_t}$ is the restriction of a linear surjection $\mathbb{R}^n \rightarrow \mathbb{R}^k$ for all t .

Theorem 4.2.4 (Cellular Approximation Theorem)

Let X, Y be CW complexes. Let $f : X \rightarrow Y$ be a map. Then the following are true.

- f is homotopic to a cellular map (called its cellular approximation).
- If f is homotopic to a cellular map f' and f is already cellular for some subcomplex $A \subseteq X$, then the homotopy to its cellular approximation can be taken such that $f'|_A = f|_A$.

Example 4.2.5

Let $0 \leq n < m$. Then $\pi_n(S^m) = 0$.

Proof

Let $[f] \in \pi_n(S^m)$. By the above theorem, we know that $f : S^n \rightarrow S^m$ is homotopic to a cellular map, where S^n is given the CW structure of a 0-cell and an n -cell. But the n -skeleton of S^m only consists of its 0-cell. Hence the cellular approximation of f is null-homotopic. Hence $[f] = 0$. ■

Example 4.2.6

$S^\infty = \operatorname{colim}_{n \in \mathbb{N}} S^n$ is weakly contractible and contractible.

Proof

Let $n \in \mathbb{N}$. Let $[f] \in \pi_n(S^\infty)$. Then $f : S^n \rightarrow S^\infty$ has a compact image since S^n is compact. By definition of the colimit, we have that $\{S^n \mid n \in \mathbb{N}\}$ is an open cover of S^∞ , hence the image of f lies in $S^{n_1}, \dots, S^{n_k}$ for some $n_1, \dots, n_k \in \mathbb{N}$. Let $m = \max\{n_1, \dots, n_k, n + 1\}$. Since the colimit is a direct limit and each map is an inclusion, we have that f factors into the following:

$$\begin{array}{ccc} S^n & \xrightarrow{f} & S^\infty \\ & \searrow g & \nearrow \\ & S^m & \end{array}$$

But $m > n$ since $m \geq n + 1$, and we know that $\pi_n(S^m) = 0$ for $m > n$. Hence g is null homotopic, and so is f . Since S^∞ is a CW complex, by Whitehead's theorem, S^∞ is contractible. ■

Theorem 4.2.7 (Relative Cellular Approximation) Any map $f : (X, A) \rightarrow (Y, B)$ between pairs of CW-complexes has a cellular approximation.

Corollary 4.2.8 Let $A \subset X$ be CW-complexes and suppose that all cells $X \setminus A$ have dimension larger than n . Then (X, A) is n -connected.

Corollary 4.2.9 Let X be a CW complex and let X^n be its n -skeleton. Then (X, X^n) is n -connected. Moreover, the inclusion $X^n \hookrightarrow X$ induces an isomorphism

$$\pi_k(X^n) \rightarrow \pi_k(X)$$

for $0 \leq k < n$ and a surjection for $k = n$.

4.3 CW Approximations

Definition 4.3.1 (CW Approximation) Let X be a space. A CW approximation of X is a weak homotopy equivalence $f : Z \rightarrow X$ where Z is a CW complex.

The goal of this section is that every space has a CW approximation. The given homotopy equivalence makes this notion powerful because this means that for any space X , there exists a CW-complex such that X and Z are homotopy equivalent, and moreover, has isomorphic homotopy, homology and cohomology groups.

Definition 4.3.2 (CW Model) Let (X, A) be a non-empty pair of CW-complexes. An n -connected CW model of (X, A) is an n -connected CW pair (Z, A) together with a map $f : Z \rightarrow X$ with $f|_A = \operatorname{id}_A$ such that

$$f_* : \pi_i(Z) \rightarrow \pi_i(X)$$

is an isomorphism for $i > n$ and an injection for $i = n$ for any choice of base point.

Theorem 4.3.3 For any non-empty pair (X, A) of CW-complexes, there exists an n -connected model (Z, A) . Moreover, Z can be built from A by attaching cells of dimension greater than n .

Theorem 4.3.4 Every pair of spaces (X, A) has a CW approximation. Such a CW approximation is unique up to homotopy equivalence.

4.4 Eilenberg-MacLane Spaces

Definition 4.4.1 (Eilenberg-MacLane Space)

Let G be a group and $n \in \mathbb{N}$. We say that a space X is an Eilenberg-MacLane space of type $K(G, n)$ if

$$\pi_k(X) = \begin{cases} K(G, n) & \text{if } k = n \\ 0 & \text{otherwise} \end{cases}$$

We often denote this space X directly by $X = K(G, n)$.

Proposition 4.4.2

Let G be a group. Then there exists a $K(G, 1)$ -CW complex.

Proposition 4.4.3

Let G be an abelian group and $n \geq 2$. Then there exists a $K(G, n)$ -CW complex. Moreover, it is unique up to homotopy equivalence.

5 Properties of the Homotopy Groups

5.1 The Homotopy Excision Theorem

Theorem 5.1.1 (The Homotopy Excision Theorem)

Let X be a CW-complex and A, B be sub complexes such that $X = A \cup B$ and $A \cap B \neq \emptyset$. If $(A, A \cap B)$ is m -connected and $(B, A \cap B)$ is n -connected for $m, n \geq 0$, then the map

$$\pi_k(\iota) : \pi_k(A, A \cap B) \rightarrow \pi_k(X, B)$$

induced by the inclusion $\iota : (A, A \cap B) \rightarrow (X, B)$ is an isomorphism for $0 \leq k < m + n$ and a surjection for $k = m + n$.

Proposition 5.1.2 Let (X, A) be a pair of r -connected CW complexes and let A be s -connected. Then the map

$$p_* : \pi_k(X, A) \rightarrow \pi_k(X/A)$$

induced by the quotient map $p : X \rightarrow X/A$ is an isomorphism for $0 \leq k \leq r + s$ and a surjection for $k = r + s + 1$.

5.2 Freudenthal Suspension Theorem

Theorem 5.2.1 (Freudenthal Suspension Theorem)

Let X, Y be CW complexes. Suppose that Y is n -connected. Then the induced map

$$\Sigma : [X, Y] \rightarrow [\Sigma X, \Sigma Y]$$

Then the following are true.

- If $\dim(X) \leq 2n$, then the map is an isomorphism.
- If $\dim(X) = 2n + 1$, then the map is a surjection.

Corollary 5.2.2 (Freudenthal Suspension Theorem for Homotopy Groups)

Let X be an n -connected CW complex. Then the following are true regarding the suspension map

$$\Sigma : \pi_k(X) \rightarrow \pi_{k+1}(\Sigma X)$$

- It is an isomorphism for $0 \leq k \leq 2n$.
- It is a surjection for $k = 2n + 1$.

Corollary 5.2.3

Let X be an n -connected CW complex. Then the following are true.

- $\pi_k(\Sigma X) \cong \pi_{k-1}(X)$ for $1 \leq k \leq 2n + 1$.
- ΣX is $(n + 1)$ -connected.

Corollary 5.2.4

Let X be an n -connected CW complex. Let $k \in \mathbb{N}$. Then the sequence

$$\pi_k(X) \rightarrow \pi_{k+1}(\Sigma X) \rightarrow \cdots \rightarrow \pi_{k+r}(\Sigma^r X) \rightarrow \cdots$$

of abelian groups satisfy the following.

- If $k \leq 2n$, then the sequence is constant.
- If $k > 2n$, then the sequence is constant after $r = k - 2n$.

Proof

Suppose that $k \leq 2n$. By Freudenthal suspension theorem, if $k \leq 2n$ then $\pi_k(X) \cong \pi_{k+1}(\Sigma X)$. Then ΣX is $(n + 1)$ -connected hence $\pi_{k+1}(\Sigma X) \cong \pi_{k+2}(\Sigma^2 X)$ is an isomorphism since $k + 1 \leq$

$2n + 2$. More generally, for $r \in \mathbb{N}$, $\Sigma^r X$ is $(r + n)$ -connected hence

$$\pi_{k+r}(\Sigma^r X) \cong \pi_{k+r+1}(\Sigma^{r+1} X)$$

is an isomorphism since $k + r \leq 2n + 2r$.

Now if $k > 2n$, then there exists $r \in \mathbb{N}$ such that $k + r \leq 2n + 2r$. Such an r is given by say $k - 2n$. Then by Freudenthal suspension theorem,

$$\pi_{k+r}(\Sigma^r X) \cong \pi_{k+r+1}(\Sigma^{r+1} X)$$

is an isomorphism. More generally, for $m \in \mathbb{N}$, $\Sigma^{r+m} X$ is $(r + m + n)$ -connected hence

$$\pi_{k+r+m}(\Sigma^{r+m} X) \cong \pi_{k+r+m+1}(\Sigma^{r+m+1} X)$$

is an isomorphism since $k + r + m \leq 2n + 2r + 2m$. ■

Corollary 5.2.5

We have

$$\pi_n(S^n) \cong \mathbb{Z}$$

for all $n \in \mathbb{N}$.

Proof

Choose $X = S^1$ in Freudenthal suspension theorem. It is 0-connected and so we have a surjection $\pi_1(S^1) \rightarrow \pi_2(S^2)$ is a surjection by Freudenthal suspension theorem. Since $\pi_1(S^1) \cong \mathbb{Z}$, we conclude that either $\pi_2(S^2) \cong \mathbb{Z}$ or $\pi_2(S^2) \cong \mathbb{Z}/d\mathbb{Z}$ for some $d \geq 1$. But since $H_2(S^2) \cong \mathbb{Z}$, this implies that a map $S^2 \rightarrow S^2$ of degree i for all $i \in \mathbb{Z}$ exists. But maps of different degrees are not homotopic. Hence $\pi_2(S^2) \cong \mathbb{Z}$.

Now choose $X = S^2$ in Freudenthal suspension theorem. It is 1-connected. By corollary 1.2.3, we deduce that the sequence

$$\pi_2(S^2) \rightarrow \pi_3(S^3) \rightarrow \cdots \rightarrow \pi_n(S^n) \rightarrow \cdots$$

is constant. Hence $\pi_n(S^n) \cong \mathbb{Z}$. ■

Example 5.2.6

Let $n \in \mathbb{N}$. Then we have

$$\pi_n \left(\bigvee_{i \in I} S^n \right) \cong \bigoplus_{i \in I} \pi_n(S^n) \cong \mathbb{Z}^{\oplus |I|}$$

given by factoring a map $S^n \rightarrow \bigvee_{i \in I} S^n$ into its components.

Proof

Suppose first that I is finite. Notice that $\prod_{i \in I} S^n$ has cells in dimensions a multiple of n . Hence $(\prod_{i \in I} S^n \mid \bigvee_{i \in I} S^n)$ is $(2n - 1)$ -connected. By the long exact sequence of relative homotopy groups, we deduce an isomorphism

$$\pi_n \left(\bigvee_{i \in I} S^n \right) \cong \pi_n \left(\prod_{i \in I} S^n \right) \cong \bigoplus_{i \in I} \pi_n(S^n)$$

Now if I is not finite, consider the map $\bigoplus_{i \in I} \pi_n(S^n) \rightarrow \pi_n \left(\bigvee_{i \in I} S^n \right)$ given by composition $S^n \rightarrow S^n \hookrightarrow \bigvee_{i \in I} S^n$. This map is surjective since any map $S^n \rightarrow \bigvee_{i \in I} S^n$ has compact image because S^n is compact so that this map factors through a finite wedge, and we have proven such a map comes from $\bigoplus_{i \in I} \pi_n(S^n)$ in the finite case. Similarly, f lies in the image of the map and is null homotopic, the null homotopy must lie in a finite wedge, and we have proven in the finite case that the f must be null homotopic along each factor in $\bigoplus_{i \in I} \pi_n(S^n)$. ■

Example 5.2.7

Let $n \geq 2$. Then we have

$$\pi_n(S^1 \vee S^n) \cong \mathbb{Z}[t, t^{-1}]$$

Example 5.2.8

Let X be the CW complex given by attaching an $(n+1)$ -cell by the attaching map $\varphi : S^n \rightarrow \bigvee_{i \in I} S^n$. Then we have

$$\pi_k(X) \cong \begin{cases} \frac{\bigoplus_{i \in I} \pi_n(S^n)}{\langle [\varphi] \rangle} & \text{if } n = k \\ 0 & \text{otherwise} \end{cases}$$

5.3 The Homotopy Groups of a Fiber Bundle

Theorem 5.3.1 Let $p : E \rightarrow B$ be a fiber bundle. Let $A \subseteq B$. Let $y_0 \in E$ and $p(y_0) = x_0$. Then there is an isomorphism

$$\pi_n(E, p^{-1}(A), y_0) \cong \pi_n(B, A, x_0)$$

given by the induced map p_* for all $n \geq 2$.

5.4 Weak Equivalences and Homological Isomorphisms

Proposition 5.4.1 Let X, Y be spaces and let $f : X \rightarrow Y$ be a weak homotopy equivalence. Then f induces isomorphisms

$$f_* : H_n(X; G) \xrightarrow{\cong} H_n(Y; G) \quad \text{and} \quad f^* : H^n(Y; G) \xrightarrow{\cong} H^n(X; G)$$

for any group G and all $n \in \mathbb{N}$.

This theorem shows that the higher homotopy groups is not a weaker invariant than homology and cohomology. Indeed, the theorem states that if the all homotopy groups are isomorphic, then all their (co)homology groups will be isomorphic.

5.5 The Hurewicz Homomorphism

Theorem 5.5.1 (Hurewicz's Homomorphism) Let X be a path connected space. Then for any $n \in \mathbb{N}$, there is a group homomorphism

$$h_n : \pi_n(X) \rightarrow H_n(X)$$

called the Hurewicz homomorphism, defined as follows. Let $[u_n] \in H_n(S^n)$ be a canonical generator. Then $h_n([f]) = f_*(u_n)$.

Theorem 5.5.2 (Hurewicz's Theorem) Let X be a space. Then the following are true regarding Hurewicz's homomorphism.

- Let $n \geq 2$. If X is $(n-1)$ -connected, then $\tilde{H}_k(X) = 0$ for all $0 \leq k < n$. Moreover, the Hurewicz homomorphism

$$h_n : \pi_n(X) \rightarrow H_n(X)$$

is an isomorphism. Moreover, h_{n+1} is a surjection.

- Let $n = 1$, then Hurewicz's homomorphism induces an isomorphism

$$\bar{h}_1 : \pi_1(X)^{\text{ab}} \rightarrow H_1(X)$$

Theorem 5.5.3 (Relative Hurewicz's Homomorphism) Let (X, A) be a pair of spaces. Then for

any $n \geq 1$, there is a group homomorphism

$$h_n : \pi_n(X, A) \rightarrow H_n(X, A)$$

called the relative Hurewicz homomorphism, defined as follows. Let $[u_n] \in H_n(S^n, \partial S^n)$ be a canonical generator. Then $h_n([f]) = f_*(u_n)$.

Theorem 5.5.4 (Relative Hurewicz's Theorem) Let (X, A) be a pair of spaces. Let $n \geq 2$. If X and A are path connected and (X, A) is $(n - 1)$ -connected, then $H_k(X, A) = 0$ for all $0 \leq k < n$. Moreover, the Hurewicz homomorphism

$$h_n : \pi_n(X, A, x_0) \rightarrow H_n(X, A)$$

is an isomorphism.

Theorem 5.5.5 (Naturality of Hurewicz's Homomorphism) Let (X, x_0) and (Y, y_0) be pointed spaces and let $f : (X, x_0) \rightarrow (Y, y_0)$ be a map. Then the following diagram is commutative:

$$\begin{array}{ccc} \pi_k(X, x_0) & \xrightarrow{\pi_k(f)} & \pi_k(Y, y_0) \\ h_k \downarrow & & \downarrow h_k \\ H_k(X) & \xrightarrow{f_*} & H_k(Y) \end{array}$$

where h is the Hurewicz homomorphism. Moreover, a similar diagram is also commutative for the relative Hurewicz homomorphism.

The connection between the homotopy groups and the homology groups begs the question of whether there is a relationship between the homotopy groups and cohomology groups that is not implicit by the relation between homology and cohomology. This is answered in Stable Homotopy Theory, when we introduced Brown's representability theorem.

6 H-Spaces

6.1 H-Spaces and the Hopf Algebra Structure on Cohomology

Definition 6.1.1 (H-Spaces)

Let X be a space. We say that X is a H-space if there exists a map $\mu : X \times X \rightarrow X$ and a point $e \in X$ such that the maps $x \mapsto \mu(x, e)$ and $x \mapsto \mu(e, x)$ are homotopic to the identity map relative to the set $\{e\}$.

Proposition 6.1.2

Let X be a CW complex. Let $\mu : X \times X \rightarrow X$ be a map. Let $e \in X$ be a point. Then the following are equivalent.

- (X, μ, e) is a H-space.
- The maps $x \mapsto \mu(x, e)$ and $x \mapsto \mu(e, x)$ are freely homotopic to the identity.
- The maps $x \mapsto \mu(x, e)$ and $x \mapsto \mu(e, x)$ are the identity.

Proposition 6.1.3

Let X be a H-space. Let Y be a pointed space. If X and Y are pointed homotopy equivalent, then Y is a H-space.

Example 6.1.4

Let G be a topological group. Then G is a H-space.

Notice that H-spaces do not have the associative requirement for μ . Thus S^7 is a H-space even though it is not a group. Also, there are no inverse requirements like a group.

Example 6.1.5

$\mathbb{CP}^\infty$ is a H-space.

- A point in $\mathbb{CP}^\infty$ is an equivalence class of sequences $(a_n)_{n \in \mathbb{N}}$ of complex numbers all but finitely many zero. The multiplicative structure is then given by considering them as coefficients of a polynomial in $\mathbb{C}[x]$ where $(a_n)_{n \in \mathbb{N}}$ is the element $\sum_{i=0}^{\infty} a_i x^i$.

Example 6.1.6

Let $n \in \mathbb{N}$. Then $\mathbb{CP}^n$ is not an H-space.

Proof

Recall that the cohomology of complex projective space is given by

$$H^*(\mathbb{CP}^n; k) = \frac{k[x]}{(x^{n+1})}$$

where k is a field of characteristic 0 and $\deg(x) = 2$. We recall the proof that $k[x]/(x^{n+1})$ is not a graded Hopf algebra.

Suppose that it is a Hopf-algebra. Then we must have $\Delta(x) = x \otimes 1 + 1 \otimes x$ by grading restrictions. This gives us

$$0 = \Delta(x^{n+1}) = \sum_{i=0}^{n+1} \binom{n+1}{i} x^i \otimes x^{n+1-i} = \sum_{i=1}^{n-1} \binom{n+1}{i} x^i \otimes x^{n+1-i}$$

By degree considerations, we conclude that $\binom{n+1}{i}$ for $1 \leq i \leq n-1$ which is a contradiction. ■

Example 6.1.7

Let X be a pointed space. Then ΩX is a H-space.

Lemma 6.1.8

Let X be a H-space. Let Y be a pointed space. Then $[Y, X]_*$ has multiplication and identity.

Proposition 6.1.9

Let X be a H-space with multiplication given by μ . Let R be a commutative ring. Suppose that the following are true.

- $H^0(X; R) \cong R$.
- $H^n(X; R)$ is a finitely generated free R -module for each $n \in \mathbb{N}$.

Then $H^*(X; R)$ is a graded commutative Hopf algebra whose comultiplication is induced by μ^* .

6.2 Homotopy Associative H-Space and the Pontryagin Product

Definition 6.2.1 (Homotopy Associative H-Space)

Let X be a H-space. We say that X is homotopy associative if the two maps

$$\mu(-, \mu(-, -)) \quad \text{and} \quad \mu(\mu(-, -), -)$$

are homotopic.

Lemma 6.2.2

Let X be a space. Then X is a homotopy associative H-space if and only if X is a monoid object in $\mathbf{hTop}$.

Lemma 6.2.3

Let X be a homotopy associative H-space. Let Y be a space. Then $[Y, X]_*$ is a monoid.

Definition 6.2.4 (Pontryagin Product)

Let X be a homotopy associative H-space. Let R be a commutative ring. Define the Pontryagin product of X to be the composite

$$H_*(X; R) \otimes_R H_*(X; R) \xrightarrow{\times} H_*(X \times X; R) \xrightarrow{\mu_*} H_*(X; R)$$

where the first map is the cross product given by the Kunneth theorem.

Proposition 6.2.5

Let X be a homotopy associative H-space. Then the Pontryagin product gives a ring structure to the direct sum

$$H_*(X; R) = \bigoplus_{n=0}^{\infty} H_n(X; R)$$

Note: coalgebra structure on homology makes it a Hopf algebra. Also, $H_*(X; R)$ is the Hopf dual of $H^*(X; R)$ since their multiplicative / comultiplicative structure is induced by the same map $\mu : X \times X \rightarrow X$.

6.3 Group Like H-Spaces

Definition 6.3.1 (Group Like H-Spaces)

Let X be a H-space. We say that X is group like if the following are true.

- X is homotopy associative.
- There exists a map $j : X \rightarrow X$ such that $\mu \circ (\text{id}, j)$ and $\mu \circ (j, \text{id})$ are homotopic to the identity map.

Lemma 6.3.2

Let X be a space. Then X is a group like H-space if and only if X is a group object in $\mathbf{hTop}$.

Lemma 6.3.3

Let X be a group like H-space. Let Y be a space. Then $[Y, X]$ is a group.

Proposition 6.3.4

Let X be homotopy associative H-space. Then X is group like if and only if the map $(x, y) \mapsto (x, xy)$ is a homotopy equivalence.

Definition 6.3.5 (Homotopy Commutative H-Space)

Let X be a H-space. We say that X is homotopy commutative if the two maps

$$(x, y) \mapsto \mu(x, y) \quad \text{and} \quad (x, y) \mapsto \mu(y, x)$$

are homotopic.

Lemma 6.3.6

Let X be a space. Then X is a group like H-space if and only if X is an abelian group object in $\mathbf{hTop}$.

6.4 Co-H-Space

Definition 6.4.1 (Co-H-Spaces)

Let X be a space. We say that X is a co-H-space if there exists a map $c : X \rightarrow X \vee X$ and a point $e \in X$ such that the maps $p_1 \circ c$ and $p_2 \circ c$ are homotopic to the identity map relative to the set $\{e\}$, where $p_1, p_2 : X \vee X \rightarrow X$ are the pinch maps.

Definition 6.4.2 (Homotopy Co-Associative Co-H-Spaces)

Let X be a co-H-space. We say that X is homotopy co-associative if the two maps

$$(c \vee \text{id}_X) \circ c \quad \text{and} \quad (\text{id}_X \vee c) \circ c$$

Definition 6.4.3 (Cogroup Like Co-H-Spaces)

Let X be a co-H-space. We say that X is cogroup like if the following are true.

- X is homotopy co-associative co-H-space.
- There exists a map $j : X \rightarrow X$ such that $(\text{id}_X \vee j) \circ c$ and $(j \vee \text{id}_X) \circ c$ are homotopic to the identity map.

Definition 6.4.4 (Homotopy Co-Commutative Co-H-Spaces)

Let X be a co-H-space. We say that X is homotopy co-commutative if there is a homotopy

$$c \simeq T \circ c$$

where $T : X \vee X \rightarrow X \vee X$ is the swap map.

Example 6.4.5

Let $n \in \mathbb{N}$.

- If $n \geq 1$, then S^n is a group like co-H-space.
- If $n \geq 2$, then S^n is a homotopy co-commutative co-H-space.

6.5 Group Structure on the Set of Maps

Definition 6.5.1 (Group Structure on the Set of Maps)

Let X be a pointed space. Let Y be a group like H-space. Define the group structure on $[X, Y]_*$ by

Proposition 6.5.2

Let X be a pointed space. Let Y be a group like H-space. Then the following are true.

- $[X, Y]_*$ is a group.
- If Y is homotopy commutative, then $[X, Y]_*$ is abelian.

Definition 6.5.3 (Group Structure on the Set of Maps)

Let X be a cogroup like co-H-space. Let Y be a pointed space. Define the group structure on $[X, Y]_*$ by

Proposition 6.5.4

Let X be a cogroup like co-H-space. Let Y be a pointed space. Then the following are true.

- $[X, Y]_*$ is a group.
- If X is homotopy co-commutative, then $[X, Y]_*$ is abelian.

Proposition 6.5.5

Let X be a pointed space. Then the bijection

$$\pi_n(X) \cong [S^n, X]_*$$

is a group isomorphism for all $n \geq 1$.

6.6 Group Isomorphism on the Suspension-Loopspace Adjunction

Definition 6.6.1 (Group Structure on Loop Spaces)

Let X be a space. Define a group structure on ΩX as follows. Let $\cdot : \Omega X \times \Omega X \rightarrow \Omega X$ be defined as the concatenation: $(f, g) \mapsto f \cdot g$.

Proposition 6.6.2

Let X be a pointed space. Then the following are true.

- ΩX is a group like H-space.
- $\Omega^2 X$ is group like homotopy commutative H-space.

Example 6.6.3

Let X, Y be pointed space. The group structure on $[X, \Omega Y]_*$ is given by the map $+$: $[X, \Omega Y]_* \times [X, \Omega Y]_* \rightarrow [X, \Omega Y]_*$ defined by

$$([f], [g]) \mapsto [f + g]$$

where $f + g : X \rightarrow \Omega Y$ is the map $(f + g)(x) = f(x) \cdot g(x)$.

Definition 6.6.4 (Cogroup Structure on Suspensions)**Proposition 6.6.5**

Let X be a pointed space. Then the following are true.

- ΣX is a cogroup like H-space.
- $\Sigma^2 X$ is a cogroup like homotopy cocomutative H-space.

Example 6.6.6

Explicit description of cogroup structure on $[\Sigma X, Y]_*$.

Example 6.6.7

This is also a way to see why $\pi_n(X) \cong [S^n, X]_*$.

Proposition 6.6.8

Let X, Y be a pointed space. Then the natural bijection

$$[\Sigma X, Y]_* \cong [X, \Omega Y]_*$$

is a group isomorphism.

Proposition 6.6.9

Let X be a pointed space. Then there is a group isomorphism

$$\pi_n(X) \cong \pi_{n-1}(\Omega X)$$

for all $n \in \mathbb{N}^+$.

Proof

By the suspension-loopspace adjunction, we have a natural bijection

$$\pi_n(X) = [S^n, X]_* \cong [S^{n-1}, \Omega X]_* = \pi_{n-1}(\Omega X)$$

It remains to show that this map is a group isomorphism. ■

7 The Free Monoid of a Space

7.1 The James Construction

Definition 7.1.1 (James Reduced Product)

Let (X, e) be a pointed space.

- Define $J_n(X) = \frac{X^n}{\sim}$ where $(x_1, \dots, x_{i-1}, e, x_{i+1}, \dots, x_n) \sim (x_1, \dots, x_i, e, x_{i+2}, \dots, x_n)$.
- Define the James reduced product of (X, e) to be

$$J(X) = \text{colim}_{n \in \mathbb{N}} J_n(X)$$

Lemma 7.1.2

Let (X, e) be a CW complex where e is a 0-cell. Then $J(X)$ is a CW complex.

Example 7.1.3

Let R be a commutative ring. Then there are graded Hopf algebra isomorphisms

$$H^*(J(S^n); R) \cong \begin{cases} \Gamma_R[x] & \deg(x) = n & \text{if } n \text{ is even} \\ \Gamma_R[x] \otimes \Lambda(R \cdot y) & \deg(x) = 2n, \deg(y) = n & \text{if } n \text{ is odd} \end{cases}$$

where

$$\Gamma_R[x] = \frac{R\langle x_k \mid k \in \mathbb{N} \rangle}{\left\langle x_i x_j = \binom{i+j}{i} x_{i+j} \mid i, j \in \mathbb{N} \right\rangle}$$

Proposition 7.1.4

Let X be a connected CW complex. Let R be a ring. Suppose that $H_*(X; R)$ is a free R -module. Then there is an isomorphism

$$H_*(J(X); R) \cong T(\tilde{H}_*(X; R))$$

Let (X, x_0) be a pointed space. Recall that the suspension of X is defined to be

$$\Sigma X = \frac{X \times I}{X \times \partial I \cup \{x_0\} \times I}$$

Elements of ΣX look like (x, t) where $x \in X$ and $t \in I$. Since $(x, 0) \sim (x, 1)$ in the quotient, the map $t \mapsto (x, t)$ is a well defined loop in ΣX , hence an element of $\Omega \Sigma X$.

Example 7.1.5

Let $n \in \mathbb{N}$. Then there is a graded ring isomorphism

$$H_*(J(S^n); \mathbb{Z}) \cong \mathbb{Z}[x]$$

where $\deg(x) = n$.

Proposition 7.1.6

Let (X, e) be a pointed connected CW complex where e is a 0-cell. Let $\lambda : J(X) \rightarrow \Omega \Sigma X$ be the map defined by $\lambda(x) = (t \mapsto (x, t))$. Then λ is a homotopy equivalence.

7.2 The Infinite Symmetric Product

Definition 7.2.1 (Infinite Symmetric Product)

Let (X, e) be a pointed space.

- Define $SP_n(X) = \frac{X^n}{S^n}$ the orbit space, where S^n acts on X^n by $\sigma \cdot (x_1, \dots, x_n) = (x_{\sigma(1)}, \dots, x_{\sigma(n)})$.

- Define the infinite symmetric product of X to be

$$SP(X) = \operatorname{colim}_{n \in \mathbb{N}} SP_n(X)$$

where the maps $SP_n(X) \hookrightarrow SP_{n+1}(X)$ is given by inclusion: $(x_1, \dots, x_n) \mapsto (x_1, \dots, x_n, e)$.

Lemma 7.2.2

Let (X, e) be a pointed space. Then there is a homeomorphism

$$SP(X) \cong \operatorname{colim}_{n \in \mathbb{N}} \frac{J_n(X)}{S_n}$$

where S_n acts on $J_n(X)$ by $\sigma \cdot (x_1, \dots, x_n) = (x_{\sigma(1)}, \dots, x_{\sigma(n)})$.

Theorem 7.2.3 (Dold-Thom Theorem)

Let (X, e) be a pointed space. Then there are isomorphisms

$$\tilde{H}_n(X) \cong \pi_n(SP(X), e)$$

for each $n \in \mathbb{N}$.

Proposition 7.2.4

Let (X, e) be a pointed space. Then we have a factorization

$$\begin{array}{ccc} \pi_n(X, e) & \xrightarrow{\operatorname{incl}_*} & \pi_n(SP(X), e) \\ & \searrow h_n & \nearrow \cong \\ & \tilde{H}_n(X) & \end{array}$$

where h is the Hurewicz homomorphism.

8 Products between Homotopy Groups

8.1 The Samelson Product

Definition 8.1.1 (Samelson Product)

Let X, Y be spaces. Let G be a group like H -space. Let $f : X \rightarrow G$ and $g : Y \rightarrow G$ be maps. Define the Samelson product $[f, g] : X \wedge Y \rightarrow G$ by

$$[f, g] = [-, -] \circ (f \wedge g)$$

where $[-, -]$ is the commutator map on G .

Note that the Samelson product is well defined because the commutator map $[-, -] : G \times G \rightarrow G$ is trivial on $G \wedge G$ and so it descends to a map $G \wedge G \rightarrow G$.

Proposition 8.1.2

Let G be a group like H -space that is 0-connected. Then the following are true.

- The map $\pi_n(G) \times \pi_m(G) \rightarrow \pi_{n+m}(G)$ given by $([f], [g]) \mapsto [f, g]$ is a $\mathbb{Z}$ -bilinear map.
- Let $[f] \in \pi_n(G)$ and $[g] \in \pi_m(G)$. Then we have

$$[f, g] = (-1)^{nm+1}[g, f]$$

Proposition 8.1.3 (The Super Jacobi Identity)

Let G be a group like H -space that is 0-connected. Let $[f] \in \pi_n(G)$, $[g] \in \pi_m(G)$ and $[h] \in \pi_k(G)$. Then we have

$$(-1)^{nk}[f, [g, h]] + (-1)^{nm}[g, [h, f]] + (-1)^{mk}[h, [f, g]] = 0$$

Corollary 8.1.4

Let G be a group like H -space that is 0-connected. Then $\pi_*(G)$ is a super Lie algebra.

8.2 The Whitehead Product

Definition 8.2.1 (The Whitehead Product)

Let X be a pointed space. Define the Whitehead product on X to be the map $[-, -]_{\text{Wh}} : \pi_n(X) \times \pi_m(X) \rightarrow \pi_{n+m-1}(X)$ as follows. For $[f] \in \pi_n(X)$ and $[g] \in \pi_m(X)$, the Whitehead product is given by

$$[f, g]_{\text{Wh}} = (f \vee g) \circ \phi$$

where $\phi : S^{n+m-1} \rightarrow S^n \vee S^m$ is the attaching map of the $(n+m)$ -cell in the CW complex structure of $S^n \times S^m$.

Proposition 8.2.2

Let X be a pointed space. Let τ be the isomorphism $\pi_{n+1}(X) \cong \pi_n(\Omega X)$. Let $[f] \in \pi_{n+1}(X)$ and $[g] \in \pi_{m+1}(X)$. Then we have

$$\tau([f, g]_{\text{Wh}}) = (-1)^{n-1}[\tau([f]), \tau([g])]$$

Proposition 8.2.3

Let X be a pointed space.

- The Whitehead product $[-, -]_{\text{Wh}}$ is a $\mathbb{Z}$ -bilinear map.
- Let $[f] \in \pi_n(X)$ and $[g] \in \pi_m(X)$. Then we have

$$[f, g] = (-1)^{nm}[g, f]$$

Proposition 8.2.4 (The Super Jacobi Identity)

Let X be a pointed space. Let $[f] \in \pi_n(G)$, $[g] \in \pi_m(G)$ and $[h] \in \pi_k(G)$. Then we have

$$(-1)^{(n-1)(k-1)}[f, [g, h]] + (-1)^{(n-1)(m-1)}[g, [h, f]] + (-1)^{(m-1)(k-1)}[h, [f, g]] = 0$$

Corollary 8.2.5

Let X be a pointed space. Then $\pi_{*-1}(X)$ is a super Lie algebra.

8.3 The Hilton Milnor Theorem

Theorem 8.3.1 (Hilton-Milnor Theorem)

Let X, Y be connected spaces. Then there is a weak equivalence

$$\Omega\Sigma X \times \Omega\Sigma \left(\bigvee_{i=0}^{\infty} (Y \wedge X^{\wedge i}) \right) \rightarrow \Omega\Sigma(X \vee Y)$$

Corollary 8.3.2

Let X, Y be connected spaces. Then there is a zigzag of weak equivalence

$$\Omega\Sigma(X \vee Y) \simeq \Omega\Sigma X \times \Omega\Sigma Y \times \Omega\Sigma \left(\bigvee_{i,j=1}^{\infty} (X^{\wedge i} \wedge Y^{\wedge j}) \right)$$

given by twice applying the Hilton-Milnor theorem.

Corollary 8.3.3

There is a weak equivalence

$$\Omega(\Sigma X_1 \vee \cdots \vee \Sigma X_n) \rightarrow \prod_{w \in W} \Omega\Sigma \left(X_1^{\wedge w_{x_1}} \wedge \cdots \wedge X_n^{\wedge w_{x_n}} \right)$$

where W is a hall basis for the free Lie algebra on $x_1, \dots, x_n$ and w_{x_i} is the number of x_i appearing in w and the product is the product in **CGWH**.

Part III

Topics in Algebraic Topology

9 The Eilenberg-Steenrod Axioms

We have seen that the homotopy groups, the homology groups and the cohomology groups all satisfy a functorial property. This means that they can be considered as functors from the category of spaces to the category of some algebraic structures. It is meaningful to study all of them at once, and to compare different versions of homology and cohomology.

In general, there are different parameters for (co)homology theories.

- Including “for CW Pairs” means that the theory is tailored for CW pairs. If we want a general theory for any topological spaces, we must add a new axiom so that weak equivalences (isomorphisms in homotopy groups) gives isomorphisms in homology groups. This is true for CW complexes but not for arbitrary topological spaces by Whitehead’s theorem.
- “Generalized” means that the additivity and dimension axioms are dropped. This enables wilder (co)homology theories such as (co)bordism to appear. “Ordinary” means that the dimension axiom is included. In the case that we restrict to CW complexes, this axiom ensures that all (co)homology theories of this type are isomorphic to each other.
- “Reduced” theories typically throw away the relative context in order to gain more concrete computations. There is a theorem that says that determining a generalized (co)homology theory is the “same” determining a reduced theory and vice versa.

9.1 Generalized (Co)Homology Theories for Spaces

Definition 9.1.1 (Generalized Homology Theory for Spaces)

A generalized homology theory is a collection of functors and natural transformations

$$h_n : \mathbf{Top}^2 \rightarrow \mathbf{Ab} \quad \text{and} \quad \delta_n : h_n \Rightarrow h_n \circ F$$

for each $n \in \mathbb{N}$, where F is the functor defined by $F(X, A) = (A, \emptyset)$, such that the following are true.

- Homotopy Invariance: Let (X, A) and (Y, B) be pairs of spaces. If $f \simeq g : (X, A) \rightarrow (Y, B)$ are homotopy equivalent maps, then

$$h_n(f) = h_n(g) : h_n(X, A) \rightarrow h_n(Y, B)$$

- Exactness: Let (X, A) be a pair of space. Then there exists an exact sequence

$$\cdots \longrightarrow h_{n+1}(X, A) \xrightarrow{\delta_{n+1}} h_n(A, \emptyset) \xrightarrow{h_n(i)} h_n(X, \emptyset) \xrightarrow{h_n(j)} h_n(X, A) \xrightarrow{\delta_n} h_{n-1}(A, \emptyset) \longrightarrow \cdots$$

where $i : (A, \emptyset) \rightarrow (X, \emptyset)$ and $j : (X, \emptyset) \rightarrow (X, A)$ are the inclusions maps.

- Excision: Let (X, A) be a pair of space. Let $E \subseteq X$ be a subspace of X such that $\overline{E} \subseteq A^\circ \subseteq X$, then the inclusion map induces an isomorphism

$$h_n(i) : h_n(X \setminus E, A \setminus E) \xrightarrow{\cong} h_n(X, A)$$

for all $n \in \mathbb{N}$.

- Weak Equivalence: Let (X, A) and (Y, B) be pairs of spaces. If $f : (X, A) \rightarrow (Y, B)$ is a weak equivalence, then there are isomorphisms

$$h_n(f) : h_n(X, A) \xrightarrow{\cong} h_n(Y, B)$$

for all $n \in \mathbb{N}$.

Note that the axiom for weak equivalence is not present in the definition of generalized homology theories for CW complexes. Indeed, we have just shown that weak equivalences between CW complexes are homotopy equivalences. This also shows that this definition really is redundant:

Proposition 9.1.2

Any generalized homology theory on $\mathbf{Top}^2$ determines and is determined by a generalized homology theory on $\mathbf{CW}^2$.

Definition 9.1.3 (Additive Homology Theory)

Let $h_n : \mathbf{Top}^2 \rightarrow \mathbf{Ab}$ be a generalized homology theory. We say that h_n is additive if the following axiom is satisfied.

- Additivity: If $(X, A) = \coprod_{i \in I} (X_i, A_i)$, then the direct sum of the inclusion maps

$$\bigoplus_{i \in I} h_n(X_i, A_i) \cong h_n(X, A)$$

is an isomorphism

Definition 9.1.4 (Ordinary Homology Theory)

Let $h_n : \mathbf{Top}^2 \rightarrow \mathbf{Ab}$ be a generalized homology theory. We say that h_n is ordinary if there exists an abelian group A such that the following axiom is satisfied.

- Dimension:

$$h_n(*) = \begin{cases} A & \text{if } n = 0 \\ 0 & \text{otherwise} \end{cases}$$

Theorem 9.1.5 (Eilenberg-Steenrod Uniqueness Theorem) Let $T : (h_n, \delta_n) \rightarrow (h'_n, \delta'_n)$ be a natural transformation of generalized homology theories defined on $\mathbf{CW}^2$ such that $h_n(*) \cong h'_n(*)$, then T is a natural isomorphism

$$(h_n, \delta_n) \cong (h'_n, \delta'_n)$$

Definition 9.1.6 (Generalized Cohomology Theory for Spaces)

A generalized cohomology theory is a collection of contravariant functors and natural transformations

$$h^n : \mathbf{Top}^2 \rightarrow \mathbf{Ab} \quad \text{and} \quad \delta^n : h^n \circ F \Rightarrow h^{n+1} \circ F$$

for each $n \in \mathbb{N}$, where F is the functor defined by $F(X, A) = (A, \emptyset)$, such that the following are true.

- Homotopy Invariance: Let (X, A) and (Y, B) be pairs of spaces. If $f \simeq g : (X, A) \rightarrow (Y, B)$ are homotopy equivalent, then

$$h^n(f) = h^n(g) : h^n(X, A) \rightarrow h^n(Y, B)$$

- Exactness: Let (X, A) be a pair of spaces. Then there exists an exact sequence

$$\cdots \longrightarrow h^{n-1}(A, \emptyset) \xrightarrow{\delta^{n-1}} h^n(X, A) \xrightarrow{h^n(j)} h^n(X, \emptyset) \xrightarrow{h^n(i)} h^n(A, \emptyset) \xrightarrow{\delta^n} h^{n+1}(A, \emptyset) \longrightarrow \cdots$$

where $i : (A, \emptyset) \rightarrow (X, \emptyset)$ and $j : (X, \emptyset) \rightarrow (X, A)$ are the inclusion maps.

- Excision: Let (X, A) be a pair of space. Let $E \subseteq X$ be a subspace of X such that $\overline{E} \subseteq A^\circ \subseteq X$, then the inclusion map induces an isomorphism

$$h^n(i) : h^n(X, A) \xrightarrow{\cong} h^n(X \setminus E, A \setminus E)$$

for all $n \in \mathbb{N}$.

- Weak Equivalence: Let (X, A) and (Y, B) be pairs of spaces. If $f : (X, A) \rightarrow (Y, B)$ is a weak equivalence, then there are isomorphisms

$$h^n(f) : h^n(Y, B) \xrightarrow{\cong} h^n(X, A)$$

for all $n \in \mathbb{N}$.

Definition 9.1.7 (Additive Cohomology Theories)

Let $h_n : \mathbf{Top}^2 \rightarrow \mathbf{Ab}$ be a generalized cohomology theory. We say that h_n is additive if it satisfies the following axiom.

- Additivity: Denote the inclusion map as $\iota_k : (X_k, A_k) \rightarrow \coprod_{k \in I} (X_k, A_k)$. Then the product of the induced maps of inclusions induce an isomorphism

$$\prod_{k \in I} (\iota_k)^* : h^n \left(\prod_{k \in I} (X_k, A_k) \right) \xrightarrow{\cong} \prod_{k \in I} h^n(X_k, A_k)$$

Proposition 9.1.8 Let F and G be two generalized cohomology theories for CW pairs. Let $\lambda : F \Rightarrow G$ be a natural transformation such that

$$\lambda : F(*, \emptyset) \xrightarrow{\cong} G(*, \emptyset)$$

is an isomorphism for $*$ the one-point space. Then λ is a natural isomorphism.

9.2 Reduced (Co)Homology Theory for CW Complexes

Definition 9.2.1 (Reduced Homology Theory for CW Complexes)

A reduced Homology Theory is a collection of functors

$$\tilde{h}_n : \mathbf{CW}_* \rightarrow \mathbf{Ab}$$

that satisfies the following.

- Homotopy Invariance: Let (X, x_0) and (Y, y_0) be pointed CW complexes. If $f \simeq g : (X, x_0) \rightarrow (Y, y_0)$ are homotopy equivalent, then

$$\tilde{h}_n(f) = \tilde{h}_n(g) : \tilde{h}_n(X, x_0) \rightarrow \tilde{h}_n(Y, y_0)$$

- Exactness: If (X, x_0) is a pointed CW-complex and (A, x_0) is a subcomplex, then there is an exact sequence

$$\tilde{h}_n(A, x_0) \xrightarrow{\tilde{\iota}} \tilde{h}_n(X, x_0) \xrightarrow{\tilde{p}} \tilde{h}_n(X/A, *)$$

for each $n \in \mathbb{N}$, where $\iota : A \rightarrow X$ is the inclusion and $p : X \rightarrow X/A$ is the projection.

- Suspension: Let (X, x_0) be a pointed CW complex. Then the suspension functor induces a natural isomorphism

$$\Sigma : \tilde{h}_n(X, x_0) \xrightarrow{\cong} \tilde{h}_{n+1}(\Sigma X, *)$$

Definition 9.2.2 (Additive Reduced Homology Theories)

Let $\tilde{h}_n : \mathbf{CW}_* \rightarrow \mathbf{Ab}$ be a reduced homology theory. We say that $\tilde{h}_n$ is additive if it satisfies the following axiom.

- Additivity: Let (X_k, x_k) be a collection of CW complexes for $k \in I$. Denote the inclusion map as $\iota_k : (X_k, x_k) \hookrightarrow \bigvee_{i \in I} (X_i, x_i)$. Then there is an isomorphism

$$\bigoplus_{i \in I} \tilde{h}_n(\iota_i) : \bigoplus_{i \in I} \tilde{h}_n(X_i, x_i) \xrightarrow{\cong} \tilde{h}_n \left(\bigvee_{i \in I} (X_i, x_i) \right)$$

Lemma 9.2.3 Let $\tilde{h}_n : \mathbf{CW}_* \rightarrow \mathbf{Ab}$ be a reduced homology theory. Then

$$\tilde{h}_n(*) = 0$$

Proposition 9.2.4 Let $h_n : \mathbf{CW}^2 \rightarrow \mathbf{Ab}$ be a generalized homology theory for CW complexes. Define a collection of functors

$$\tilde{h}_n : \mathbf{CW}_* \rightarrow \mathbf{Ab}$$

by $(X, x_0) \mapsto \tilde{h}_n(X, x_0) = h_n(X, \{x_0\})$. Then $\tilde{h}_n$ defines a reduced homology theory for CW complexes.

Proposition 9.2.5 Let $\tilde{h}_n : \mathbf{CW}_* \rightarrow \mathbf{Ab}$ be a reduced homology theory for CW complexes. Define a collection of functors

$$\tilde{h}_n : \mathbf{CW}^2 \rightarrow \mathbf{Ab}$$

by $(X, A) \mapsto \tilde{h}_n(X/A, * = A)$ and a collection of natural transformations

$$\delta_n : h_n(X, Y) \rightarrow (Y, \emptyset)$$

by ??? Then h_n and δ_n defines a generalized homology theory for CW complexes.

Proposition 9.2.6 Any generalized homology theory for CW complexes determines and is determined by a reduced homology theory for CW complexes.

Thus we have showed that under the conditions of all the given axioms of all such homology theories, we really just have the same thing:

$$\begin{array}{ccccc} \text{Reduced Homology Theory} & \rightleftharpoons & \text{Generalized Homology Theory} & \rightleftharpoons & \text{Generalized Homology Theory} \\ \text{for CW Complexes} & & \text{for CW Complexes} & & \text{for Spaces} \end{array}$$

Note that such a bijection is only possible when we introduce the weak equivalence axiom for homology theories on spaces. Such an axiom in fact restricts the amount of valid homology theories, but it is only through this axiom that we can declare all such homology theories are essentially determined by one another.

Some authors indeed does not require the weak equivalence axiom to exist. In such cases the bijection between generalized homology theories and reduced homology theories can be constructed, but there is no passage between homology of CW complexes and homology of spaces in general.

Definition 9.2.7 (Reduced Cohomology Theory for CW Complexes)

A reduced cohomology theory is a collection of contravariant functors

$$h^n : \mathbf{CW}_* \rightarrow \mathbf{Ab}$$

,for each $n \in \mathbb{N}$ such that the following are true.

- Homotopy Invariance: If $f \simeq g : X \rightarrow Y$ then

$$h^n(f) = h^n(g) : h^n(X) \rightarrow h^n(Y)$$

- Exactness: If $f : A \rightarrow X$ is an inclusion of CW complexes, then there is an exact sequence

$$h^n(C_f) \xrightarrow{h^n(i)} h^n(X) \xrightarrow{h^n(f)} h^n(A)$$

where $i : X \rightarrow C_f$ is the inclusion map.

- Suspension: There is a natural isomorphism of functors:

$$\sigma : h^{n+1} \circ \Sigma \Rightarrow h^n$$

- Additivity: Let $i_k : X_k \rightarrow \bigvee_{i \in I} X_i$ denote the inclusion map. Their products induce an isomorphism

$$\prod_{k \in I} h^n(i_k) : h^n\left(\bigvee_{i \in I} X_i\right) \xrightarrow{\cong} \prod_{i \in I} h^n(X_i)$$

Lemma 9.2.8 Let $\tilde{h}_n : \mathbf{CW} \rightarrow \mathbf{Ab}$ be a reduced homology theory. Then

$$\tilde{h}_n(*) = 0$$

9.3 Brown Representability Theorem

Theorem 9.3.1 (Brown's Representability Theorem for Singular Cohomology)

Let G be an abelian group. Then there is a natural isomorphism

$$H^n(-; G) \cong [-, K(G, n)]$$

for each $n \in \mathbb{N}$.

10 Stable Phenomena

10.1 The Stable Homotopy Groups

Definition 10.1.1 (Stable Homotopy Groups)

Let X be a space. Let $n \in \mathbb{N}$. Define the n th stable homotopy groups of X to be

$$\pi_n^s(X) = \operatorname{colim}_{k \rightarrow \infty} \pi_{n+k}(\Sigma^k X)$$

This is well defined because of the Freudenthal suspension theorem, which states that the groups in the direct limit eventually stabilize. Indeed, the same theorem shows the following.

Proposition 10.1.2 Let X be a space. Then there is an isomorphism

$$\pi_n^s(X) \cong \pi_{n+1}^s(\Sigma X)$$

induced by the suspension functor between stable homotopy groups for any $n \in \mathbb{N}$.

Proof This can be seen by simply unwinding the definitions. We have that

$$\begin{aligned} \pi_{n+1}^s(\Sigma X) &= \operatorname{colim}_{k \rightarrow \infty} \pi_{n+k+1}(\Sigma^{k+1} X) \\ &= \operatorname{colim}_{k \rightarrow \infty} \pi_{n+k}(\Sigma^k X) \\ &= \pi_n^s(X) \end{aligned}$$

and so we conclude. ■

New: Graded ring structure on π_*^s .

Theorem 10.1.3 The stable homotopy groups define a collection of functors $\pi_n^s : \mathbf{CW}_* \rightarrow \mathbf{Ab}$ as follows.

- For X a pointed CW complex, $\pi_n^s(X)$ is the n th stable homotopy group of X
- For $f : X \rightarrow Y$ a map,

$$\pi_n^s(f) : \pi_n^s(X) \rightarrow \pi_n^s(Y)$$

is the image of f under the canonical map $[X, Y] \rightarrow [X, Y]^s$

Theorem 10.1.4 The stable homotopy functors $\pi_n^s : \mathbf{CW}_* \rightarrow \mathbf{Ab}$ for each $n \in \mathbb{N}$ defines a reduced homology theory.

This is untrue for the unstable homotopy groups. In particular, the fundamental group is not necessarily abelian.

For a space X , we may ask if there is a space Y such that $\pi_n^s(X) = \pi_n(Y)$. The answer is yes and in fact Y can be constructed functorially from X .

Definition 10.1.5 (The Q-Functor)

Let X be a space. Define

$$QX = \operatorname{colim}_{k \rightarrow \infty} \Omega^k \Sigma^k X$$

Lemma 10.1.6

Let X be a space. Then we have

$$\pi_n^s(X) = \pi_n(QX)$$

for all $n \in \mathbb{N}$.

Definition 10.1.7 (Delooping)

Let X, Y be spaces. We say that Y is a delooping of X if there is a weak equivalence $X \xrightarrow{\sim} \Omega Y$.

Definition 10.1.8 (Infinite Loopspace)

Let X be a space. We say that X is an infinite loopspace if there is a sequence of spaces $(X_n)_{n \in \mathbb{N}}$ with $X = X_0$ together with deloopings

$$X_n \xrightarrow{\sim} \Omega X_{n+1}$$

Lemma 10.1.9

Let X be a space. Then QX is an infinite loopspace.

10.2 Spanier Whitehead Duality

Recall that for $n \geq 2$, the set $[\Sigma^2 X, Z]$ for any spaces X and Z are abelian. Moreover, the suspension map $\Sigma : [\Sigma^2 X, Z] \rightarrow [\Sigma^3 X, \Sigma Z]$ is a group homomorphism. In particular, we can choose $Z = \Sigma^2 Y$. Now since the category **Ab** of abelian groups is cocomplete, the following inverse system

$$[\Sigma^2 X, \Sigma^2 Y] \xrightarrow{\Sigma} [\Sigma^3 X, \Sigma^3 Y] \xrightarrow{\Sigma} [\Sigma^4 X, \Sigma^4 Y] \longrightarrow \dots$$

has an inverse limit. This leads to the following definition.

Definition 10.2.1 (Set of Stable Homotopy Classes of Maps) Let X and Y be space. The set of stable homotopy classes of maps from X to Y is defined to be the abelian group

$$[X, Y]^s = \operatorname{colim}_{n \in \mathbb{N} \setminus \{0,1\}} [\Sigma^n X, \Sigma^n Y]$$

The following observation is crucial. When X and Y are pointed CW complexes, the abelian groups $[\Sigma^n X, \Sigma^n Y]$ also become stable under suspensions.

Theorem 10.2.2 Let X and Y be pointed CW complexes. Then the sequence of abelian groups,

$$[\Sigma^n X, \Sigma^n Y] \xrightarrow{\Sigma} [\Sigma^{n+1} X, \Sigma^{n+1} Y] \longrightarrow \dots$$

are isomorphic under the suspension functor for $n > \dim(X)$ and $n \geq 2$.

New: When X is compact, $[X, QY]_* = [X, Y]^s_*$.

11 Spectral Sequences in Algebraic Topology

11.1 Spectral Sequences in Topology

Theorem 11.1.1 Let X be a space. Let the following be a sequence

$$\emptyset \subset X_0 \subset X_1 \subset \cdots \subset X$$

of subspaces. Let G be an abelian group. Then the following data

- $A_{p,q} = H_{p+q}(X_p; G)$
- $E_{p,q} = H_{p+q}(X_p, X_{p-1}; G)$
- $i : H_{p+q}(X_p; G) = A_{p,q} \rightarrow H_{p+q}(X_{p+1}; G) = A_{p+1,q-1}$ (degree $(1, -1)$)
- $j : H_{p+q}(X_p; G) = A_{p,q} \rightarrow H_{p+q}(X_p, X_{p-1}; G) = E_{p,q}$ (degree $(0, 0)$)
- $k : H_{p+q}(X_p, X_{p-1}; G) = A_{p,q} \rightarrow H_{p+q-1}(X_{p-1}; G) = A_{p-1,q}$ (degree $(-1, 0)$)

defines an exact couple and hence a spectral sequence with E^1 page given by

$$E_{p,q}^1 = H_{p+q}(X_p, X_{p-1}; G)$$

where the differential $d : E_{p,q}^1 \rightarrow E_{p-1,q}^1$ is given by the composition

$$H_{p+q}(X_p, X_{p-1}; G) \xrightarrow{k} H_{p+q-1}(X_{p-1}; G) \xrightarrow{j} H_{p+q-1}(X_{p-1}, X_{p-2}; G)$$

The E_1 page of such a spectral sequence is given by

$$\begin{array}{ccccccc} \cdots & \longleftarrow & \cdots & \longleftarrow & \cdots & \longleftarrow & \cdots \\ & & H_3(X_1, X_0; G) & \longleftarrow & H_4(X_2, X_1; G) & \longleftarrow & H_4(X_3, X_2; G) \longleftarrow \cdots \\ & & & & & & \\ & & H_2(X_1, X_0; G) & \longleftarrow & H_3(X_2, X_1; G) & \longleftarrow & H_4(X_3, X_2; G) \longleftarrow \cdots \\ & & & & & & \\ & & H_1(X_1, X_0; G) & \longleftarrow & H_2(X_2, X_1; G) & \longleftarrow & H_3(X_3, X_2; G) \longleftarrow \cdots \end{array}$$

Things get interesting when we choose X to be a CW complex and we choose the filtration of X by the skeleton of X . Recall that we have the formula

$$H_{p+q}(X_p, X_{p-1}; G) \cong \begin{cases} C_p^{\text{CW}}(X; G) & \text{if } q = 0 \\ 0 & \text{otherwise} \end{cases}$$

Thus the E^1 page is only left with a chain complex at $q = 0$.

Let us also compute the derived couple of this exact couple or in other words, the E^2 page of the spectral sequence. This is more intuitive than the one thinks about on the definition of the derived couple. The $E_{p,q}^2$ slot is simply the homology of the chain complex at the (p, q) th slot. The direction of the maps of the E^2 page depends not on the choice of spectral sequence at all (In fact, the direction only depends on the page). Now in our case, the homology can be given by a known construct:

$$E_{p,q}^2 = \frac{\ker(d : H_{p+q}(X_p, X_{p-1}; G) \rightarrow H_{p+q-1}(X_{p-1}, X_{p-2}; G))}{\text{im}(d : H_{p+q+1}(X_{p+1}, X_p; G) \rightarrow H_{p+q}(X_p, X_{p-1}; G))} = H_{p+q}^{\text{CW}}(X; G)$$

Since the direction of the maps are now diagonal and when $q \neq 0$ we have $E_{p,q}^2 = 0$, all maps in E^2 are 0 and we are left with

$$H_0^{\text{CW}}(X; G) \quad H_1^{\text{CW}}(X; G) \quad H_2^{\text{CW}}(X; G) \quad H_3^{\text{CW}}(X; G) \quad \cdots$$

Theorem 11.1.2 (Leray-Serre Spectral Sequence) Let $p : E \rightarrow B$ be a Serre fibration with fibre F and path connected B . Suppose that the action of $\pi_1(B)$ on $H_*(F; G)$ is trivial. Then there is a first quadrant homological spectral sequence starting with E^2 and weakly converging to $H_\bullet(E; \mathbb{Z})$. Explicitly, there is a convergence

$$E_{p,q}^2 = H_p(B, H_1(F)) \Rightarrow H_{p+q}(E; \mathbb{Z})$$

12 Bonus?

12.1 Postnikov Towers

Definition 12.1.1 (Postnikov Towers) Let X be a path connected space. A Postnikov tower is the following commutative diagram

$$\begin{array}{ccccccc} & & X & & & & \\ & & \downarrow & \searrow & \searrow & \searrow & \\ \cdots & \longrightarrow & X_n & \xrightarrow{p_n} & X_{n-1} & \longrightarrow & \cdots \longrightarrow X_2 \xrightarrow{p_2} X_1 \xrightarrow{p_1} * \end{array}$$

such that the following are true.

- The maps $X \rightarrow X_n$ for each $n \in \mathbb{N}$ induces isomorphisms $\pi_i(X) \cong \pi_i(X_n)$ for $i \leq n$.
- $\pi_i(X_n)$ for $i > n$.
- Each $p_n : X_n \rightarrow X_{n-1}$ for $n \in \mathbb{N}$ is a fibration with fiber $K(\pi_n(X), n)$.

Theorem 12.1.2 Suppose that there is an inverse system of spaces

$$\begin{array}{ccccccc} & & \lim_{n \rightarrow \infty} X_n & & & & \\ & & \downarrow & \searrow & \searrow & \searrow & \\ \cdots & \longrightarrow & X_n & \xrightarrow{p_n} & X_{n-1} & \longrightarrow & \cdots \longrightarrow X_2 \xrightarrow{p_2} X_1 \xrightarrow{p_1} * \end{array}$$

The functor π_i for $i \in \mathbb{N}$ induces a cone in **Grp**. By definition of $\lim_{\leftarrow} \pi_i(X_n)$, there is a unique map

$$\lambda : \pi_i \left(\lim_{\leftarrow} X_n \right) \rightarrow \lim_{\leftarrow} \pi_i(X_n)$$

Then the following are true regarding λ .

- λ is surjective
- λ is injective if the maps $\pi_{i+1}(X_n) \rightarrow \pi_{i+1}(X_{n-1})$ are surjective for sufficient large n .

Proposition 12.1.3 Let X be a connected CW complex. Then there exists a Postnikov tower for X .

Proposition 12.1.4 Let X be a connected CW complex. Choose a Postnikov tower of X . Then there is a weak homotopy equivalence

$$X \simeq \lim_{\leftarrow} X_n$$

so that X is a CW approximation of $\lim_{\leftarrow} X_n$.

12.2 Reading Cell Structures from Homology Groups

Proposition 12.2.1

Let X be a simply connected CW complex. Suppose that $H_n(X)$ is a finitely generated abelian group for all $n \in \mathbb{N}$. Then there exists a CW complex Z and a cellular homotopy equivalence $f : Z \rightarrow X$ such that the cells of Z are of one of the following forms.

- A generator n -cell e_α^n corresponding to a generator α of $H_n(X)$.
- A relator $(n+1)$ -cell whose cellular boundary is equal to a multiple of the generator n -cell e_α^n if α has finite order.